

Homomorphic Signatures : A Systematization of Knowledge

Olive Chakraborty 

Université Paris-Saclay, CEA LIST, Palaiseau, France

Abstract. Homomorphic Signatures (HS) enable the authentication of data that has been processed by an untrusted party, allowing a verifier to check the correctness of a computation without access to the original signed inputs. Since their introduction, HS have evolved from algebraically restricted linear schemes to expressive non-linear and Fully Homomorphic Signature (FHS) constructions, spanning diverse cryptographic assumptions and security models.

This paper presents a Systematization of Knowledge (SoK) on homomorphic signatures. We organize existing schemes along key dimensions including functional expressiveness, underlying cryptographic primitives, security notions (selective vs. adaptive, single-key vs. multi-key), and privacy guarantees such as context hiding. This unified perspective highlights a fundamental shift from algebraic constructions toward proof-based and post-quantum designs, as well as the growing importance of Multi-Key Homomorphic Signatures (MKHS) for decentralized settings. We conclude by identifying open problems and emerging directions that must be addressed to bridge the gap between theoretical HS constructions and practical verifiable computation.

Keywords: Homomorphic Signatures · Fully Homomorphic Signatures (FHS) · Multi-Key Cryptography · Lattice-based Cryptography · Verifiable Computation · Indistinguishability Obfuscation · Post-Quantum Security · Succinctness · Context-Hiding.

1 Introduction

The modern computational landscape is being defined by a shift toward decentralized and outsourced environments, where data owners increasingly delegate complex processing tasks to untrusted third-party servers. While encryption techniques such as Fully Homomorphic Encryption (FHE) address the challenge of data privacy, they do not inherently provide integrity. Digital signatures essentially provides a mechanism for authentication thus providing integrity. However, consider the case where a server returns the result of a computation over authenticated data, and the client requires to verify that the result is indeed correct and the inputs were legitimately signed inputs. Traditional digital signatures would fail, by design they are fundamentally non-malleable. Any modification to the signed message will invalidate the signature. We observe that this technical requirement is driven by the broader paradigm of cloud computing, where the outsourcing of data storage and processing to third-party infrastructure has introduced a critical need for accountability. While the commercial sector has largely addressed data confidentiality, there remain significant open problems regarding the integrity of processed data. Because cloud providers cannot be considered fully trustworthy or immune to faults, the central question is how a client can hold the cloud accountable if a malicious deviation or a silent error occurs during processing.

E-mail: olive.chakraborty@cea.fr (Olive Chakraborty)

We argue that Homomorphic Signatures bridge this integrity gap. Homomorphic signatures (HS) are specialized primitives that remains malleable under a specific class of functions, allowing any evaluator to derive a valid signature on the output of a computation without possessing the signer’s secret key. This enables a verifier to authenticate computed results directly, without accessing or processing the original data. More generally, HS complements privacy-preserving primitives by giving integrity guarantees on top of it.

The development of HS has been largely characterised by a class of admissible functions. Initial constructions were limited to linear HS. These support additive operations over fields or groups. Linear schemes found immediate utility in network coding and statistical aggregation where integrity of linear combination is of importance. However, with the growing demand of complex analytic, there has been a significant shift towards polynomial and leveled homomorphic schemes and finally progressing towards Fully Homomorphic Signatures. These construction were largely due to lattice based techniques allowing for evaluation of circuits with increasing depth and complexity.

Alongside the growth witnessed in the functional depth a HS achieves, there has been a significant shift in allowing the inputs to be accepted from multiple users with different keys to construct a homomorphic signature. In the traditional single-key setting, all signatures are processed from a single signer. While this might work for private cloud computations, it fails to capture the use cases where there are multiple users, each with a data of its own and signing with their own key. One of the most common use case is cross-institutional medical research. This has led to emergence of multi-key HS. We emphasise that this transition introduces several technical challenges, particularly regarding the size of the evaluated signature and the complexity of the verification process.

At the same time, the predominance of lattice based constructions reflects a broader transition towards post-quantum cryptography. As classical assumptions face threats from quantum algorithms, lattices provide the necessary algebraic richness to support homomorphic operations while maintaining resistance against quantum adversaries.

Beyond algebraic constructions, recent years have witnessed a conceptual shift toward proof-based homomorphic signatures. These schemes replace explicit homomorphic structure with cryptographic proofs such as zero-knowledge arguments, functional commitments, and obfuscation, that attest to the correctness of a computation. This transition has enabled new forms of succinctness, adaptive security, and privacy (e.g., context hiding). Thus has laid the foundation for theoretical constructions of unleveled fully homomorphic signatures under strong but falsifiable assumptions.

Despite the rapid growth of the field, existing surveys have largely focused on high-level functional classifications or security definitions, often abstracting away the concrete cryptographic mechanisms underlying each scheme. This paper presents a Systematization of Knowledge (SoK) on homomorphic signatures that emphasizes construction-level understanding. We organize the literature along unifying dimensions such as functional expressiveness, cryptographic foundations, security models, privacy guarantees, and trust assumptions. We provide explicit algorithmic descriptions of representative schemes. In particular, we detail the mathematical structure of key generation, signing, verification, and homomorphic evaluation, highlighting how design choices manifest concretely in group equations, lattice relations, and proof systems.

By consolidating fragmented lines of research into a coherent framework, this SoK clarifies the design space of homomorphic signatures and identifies key open problems, including context-hiding efficiency trade-offs, correctness under approximate computation, dynamically extensible multi-key authentication, and practical post-quantum deployment. Our goal is to provide both a technical reference and a conceptual roadmap for researchers seeking to advance the theory and practice of verifiable computation in decentralized systems.

1.1 Organisation

The remainder of this survey is structured as follows: In Section 2, we establish the necessary Background and Preliminaries, covering the mathematical foundations of lattices (the SIS problem) and bilinear groups, as well as the formal security definitions for unforgeability and context hiding. Section 3 provides a detailed taxonomy of Single-Key Homomorphic Signatures, tracing the evolution from early linear schemes to polynomial and Fully Homomorphic Signatures, concluding with the recent transition toward proof-based constructions. In Section 4, we analyze the emerging field of Multi-Key Homomorphic Signatures (MKHS), categorizing foundational frameworks and modern succinct constructions that support data fusion from multiple independent signers. Finally, Section 5 explores the Future of the field, identifying emerging trends in post-quantum security and open research problems that remain to be addressed in the quest for efficient, secure, and privacy-preserving homomorphic authentication.

1.2 Related Primitives

We begin by situating HS within the broader landscape of verifiable outsourcing, distinguishing them from several primitives that share the goal of integrity but operate under different trust models.

Homomorphic signatures belong to the broader umbrella of Homomorphic Authenticators. A parallel line of research explored symmetric-key counterparts, most notably the Homomorphic Message Authentication Codes (HMACs) introduced by Backes et al. [BFR13]. While HMACs have very low computational overhead, they rely on shared secret keys. This limits us to private verification. In contrast, HS analysed in this survey, operates in the public-key setting. This allows any third party to verify the integrity of the computation, which is exactly the need in a decentralised or public auditing scenarios.

At the foundational level, Verifiable Computation (VC) [GKKR10] ensures that an untrusted server evaluates a function $f(x)$ correctly. However, we highlight a critical distinction: while VC focuses on the correctness of the process, HS certify the authenticity of the source. By binding the computation to a signer’s secret key, HS ensures that the inputs themselves are legitimate. This is a requirement that standard VC does not inherently address. The evolution of these primitives has increasingly moved toward handling complex, multi-stage data pipelines through incremental and compositional verifiable computation [Val08]. These systems allow the output of one verified computation to serve as the input for another or enable folding new computation steps into an existing proof. This concept we argue serves as a technical precursor to Multi-Key Homomorphic Signatures (MKHS) where data from multiple independent sources must be fused and verified.

We further distinguish homomorphic signatures from Functional Commitments [LRY16]. While a standard commitment allows a user to lock a value x and reveal it later, a functional commitment allows the user to commit to a dataset and later prove that $y = f(x)$ for some function f , without revealing the full dataset x . We observe that commitments often serve as a internal component of a homomorphic signatures, the signer commits to the messages, and the signature effectively acts as a proof that the committed data is authentic. This relationship is central to modern schemes like Catalano-Fiore-Tucker [CFT22], which builds homomorphic signatures directly from functional commitments to achieve better efficiency and strong context hiding. By shifting the focus from signing the data to signing a commitment of the data, these primitives allow for a more modular design where the authenticity is handled by the signature and the privacy is handled by the commitment.

The functional boundaries of HS are further clarified when compared to Functional Signatures [BGI14]. While both involve delegated signing authority, the direction of delegation differs. In a functional signature scheme, a central authority shares a portion of

its signing power, allowing another user to generate original signatures for any message that meets a specific criteria. On the other hand, HS does not allow the creation of new signatures from scratch. Instead it permits the processing and combination of existing signatures on authenticated data. We argue the vital distinction of the server being limited to the data given in the homomorphic model, while in the functional model, the user has the power to sign any message within the scope of the function. Finally, we observe a convergence between HS and the succinctness offered by Aggregate Signatures [BGLS03]. While aggregation traditionally focuses on compressing multiple signatures under the identity function, HS generalize this capability to support arbitrary computations.

This evolution towards compression has led to adoption of SNARKs [BCCT12] in modern constructions. Early schemes relied on algebraic homomorphism, where the underlying mathematical group naturally supports addition. Modern Fully Homomorphic constructions often utilize a lattice-based signature (or commitment) to authenticate the data, while a succinct proof system certifies the integrity of the evaluation. This allows to achieve the extreme succinctness required for practical, large-scale deployment.

2 Background & Preliminaries

2.1 Cryptographic Background

Many homomorphic signature constructions rely on a small number of algebraic and cryptographic primitives that recur across different schemes and security models. In particular, a large fraction of the state of the art is built either on lattice-based assumptions—most notably the Short Integer Solution (SIS) problem—or on pairing-based assumptions over bilinear groups. These foundations determine not only the security guarantees of the resulting schemes, but also their efficiency, supported function classes, and privacy properties such as context hiding. To avoid repetition and to facilitate comparison between constructions, we briefly review these two frameworks and the associated tools that are repeatedly used in homomorphic signature schemes.

2.1.1 Lattice Based Foundations : The SIS problem

Lattice based HS scheme are typically based on the hardness of the Short Integer Solution (SIS) problem. Let $q \geq 2$ be an integer modulus and let $\mathbf{A} \in \mathbb{Z}_q^{n \times m}$ be a uniformly random matrix. The SIS problem asks to find a non-zero vector $\mathbf{x} \in \mathbb{Z}^m$ such that

$$\mathbf{Ax} = 0 \pmod{q} \text{ and } \|\mathbf{x}\| \leq \beta$$

for a predefined norm bound β .

The SIS assumption states that for appropriate choices of parameters (n, m, q, β) , no probabilistic polynomial-time adversary can solve this problem with non-negligible probability. SIS is known to be at least as hard as approximating certain worst-case lattice problems, such as the Shortest Independent Vectors Problem (SIVP), within polynomial factors, thereby providing strong evidence of post-quantum security.

In homomorphic signature constructions, SIS typically arises through the unforgeability requirement: producing a valid signature without the signing key often corresponds to finding a short vector satisfying the linear relation modulo q , which directly yields an SIS solution.

A central tool in lattice-based cryptography is the use of trapdoor matrices, which enable efficient sampling of short solutions to linear equations modulo q . Two algorithms appear repeatedly across lattice-based homomorphic signature schemes: `TrapGen` and `SamplePre`.

Theorem 1. [AP11, Theorem 3.2] Let q be an integer and $m = \lceil 6n \log q \rceil$. There is a probabilistic polynomial time algorithm $\text{TrapGen}(q, n)$ that outputs a pair $(\mathbf{A}, T_A)$, where $\mathbf{A} \in \mathbb{Z}_q^{n \times m}$ and $T_A \in \mathbb{Z}^{n \times n}$, such that $\mathbf{A}$ is statistically close to a uniform matrix in $\mathbb{Z}_q^{n \times m}$ and T_A is the basis for $\Lambda_q^\perp(\mathbf{A})$ such that $\|T_A\| \leq 30\sqrt{n \log q}$.

Theorem 2. [GPV08, Theorem 4.1] There exists a probabilistic polynomial time algorithm $\text{SamplePre}(\mathbf{A}, T_A, u, \delta)$ that, given a matrix $\mathbf{A}$ and a basis T_A of $\Lambda_q^\perp(\mathbf{A})$, a parameter $\delta \geq \|T_A\| \cdot \omega(\sqrt{\log n})$ and a vector $u \in \mathbb{Z}^n$, it outputs a sample σ from the distribution that is statistically close to $\mathcal{D}_{\Lambda_q^\perp(\mathbf{A}), \delta}$.

The output distribution of σ is statistically close to a discrete Gaussian over the lattice coset defined by u , ensuring that signatures do not leak information about the trapdoor T_A . This property is essential both for security and for enabling homomorphic evaluation, since linear combinations of short pre-images remain short. In lattice-based homomorphic signature schemes, signing a message typically consists of computing a short pre-image of a message-dependent vector using SamplePre , while verification checks the corresponding linear relation modulo q . Homomorphic evaluation is enabled by the linearity of the underlying relation: linear combinations of signed messages correspond to linear combinations of their pre-images.

2.1.2 Bilinear Group Foundations

Another candidate of building homomorphic signature schemes are using pairings over bilinear groups.

Definition 1. Let $(\mathbb{G}_1, \mathbb{G}_2, \mathbb{G}_T)$ be cyclic groups of prime order p , where $\mathbb{G}_1 = \langle g \rangle$ and $\mathbb{G}_2 = \langle h \rangle$. We call $e : \mathbb{G}_1 \times \mathbb{G}_2 \mapsto \mathbb{G}_T$, a bilinear map or pairing if it satisfies the following properties:

- **Bi-linearity:** For all $a, b \in \mathbb{Z}_p$, $e(g^a, h^b) = e(g, h)^{ab}$.
- **Non-degeneracy:** $e(g, h) \neq 1_{\mathbb{G}_T}$ for generators g, h .
- **Efficient computability:** The map e can be computed in polynomial time.

The case of $\mathbb{G}_1 = \mathbb{G}_2$ is considered the Type 1 pairings where e is symmetric. When $\mathbb{G}_1 \neq \mathbb{G}_2$, then e is asymmetric and are called Type 2 pairings if there exists an efficient isomorphism from $\mathbb{G}_2 \rightarrow \mathbb{G}_1$, and Type 3 otherwise. Most modern, efficient schemes (like those after 2013) use Type 3 because they are faster and have smaller signatures.

In pairing-based homomorphic signature schemes, messages and signatures are typically encoded as group elements, and correctness is enforced through pairing equations. The bilinear property allows verification equations to naturally support linear combinations of signed messages, making bilinear groups particularly well suited for linearly homomorphic authentication. Moreover, pairings enable compact signatures and flexible security proofs, including constructions achieving adaptive unforgeability and strong context hiding in the standard model.

We observe that while the bilinear map provides the homomorphic properties, the security of the schemes relies on the hardness of certain computational problems in these groups. The most fundamental assumption is the Discrete Logarithm problem, which states that given $g^a \in \mathbb{G}$, it is infeasible to find a . However, for homomorphic signatures, one requires the Computational Diffie-Hellman (CDH) assumption to prevent forgery. Very informally, in CDH assumption, any adversary given (g, g^a, g^b) must not be able to compute g^{ab} . CDH is sufficient to construct efficient signatures in ROM, while moving to Standard model requires assumptions to handle security proof without ideal hash functions.

The Decisional Linear (DLIN) [BBS04] assumption provides a decisional challenge in pairing friendly groups where the standard Decisional DH problem is easy. The assumption

states that in a group where the standard DDH problem is easy, it is still hard to tell the difference between a linear combination of two independent DH like values and a completely random element. This problem is critical for achieving context hiding.

For schemes requiring evaluating higher degree polynomial evaluation, static assumptions like CDH or DLIN prove insufficient. However, stronger assumption like the k -Strong Diffie-Hellman (k -SDH) problem come to rescue. Informally, given a list of $k + 1$ group elements $(g, g^x, g^{x^2}, \dots, g^{x^k})$, it is hard for an adversary to produce new authenticated values that follows a specific mathematical structure. However, with the increase in the size of k , the advantage for an adversary increases. Therefore, one can achieve high-degree computation using k -SDH, albeit with lower security margins or use the previously mentioned assumptions but restricted to simpler linear operations.

2.1.3 Proofs and Commitments

Beyond constructions based on algebraic hardness assumptions of lattices or bilinear pairings, many modern homomorphic signature schemes rely on proof systems to enforce correctness and privacy. In particular, non-interactive zero knowledge proofs are used to certify that a homomorphic evaluation has been performed correctly without revealing the underlying inputs or any intermediate computation. These are very important constructions for achieving strong or complete context hiding, supporting fully HS for general circuits and multi-key evaluations. We very briefly review the proof based notions that recur throughout the literature and serve as building blocks in the constructions.

Definition 2 (NP Relation and Language). An NP relation $\mathcal{R} \subseteq \mathcal{X} \times \mathcal{W}$ consists of a set of statements $x \in \mathcal{X}$ and corresponding witnesses $w \in \mathcal{W}$ such that the membership $(x, w) \in \mathcal{R}$ can be verified in polynomial time. The triple $L_{\mathcal{R}} = (\mathcal{X}, \mathcal{W}, \mathcal{R})$ is considered a language.

NP relations provide a formal framework for expressing correctness statements about computations. In particular, they allow one to define what it means for an output to be valid with respect to some hidden input and an efficiently verifiable relation. However, NP relations alone do not specify how such correctness can be demonstrated by an untrusted party. This is where zero-knowledge proof systems come in the picture which enable efficient and privacy-preserving verification of NP statements.

Definition 3 (Non-Interactive Zero-Knowledge (NIZK) Proofs). A NIZK proof for a language $L_{\mathcal{R}}$ allows a prover to convince a verifier of a statement $x \in L_{\mathcal{R}}$, without revealing the witness $w \in \mathcal{R}$. Generally, a proof system comprises of three polynomial time algorithms:

1. $\text{crs}, \text{td}_{\text{sim}} \leftarrow \text{NIZK.Setup}(1^\lambda)$: Generates a Common Reference String and a simulation trapdoor.
2. $\pi \leftarrow \text{NIZK.Prove}(\text{crs}, x, w)$: Produces a proof π for a statement x given witness w .
3. $\{0, 1\} \leftarrow \text{NIZK.Verify}(\text{crs}, x, \pi)$: Accepts or rejects the proof.

A NIZK proof must satisfy three important properties:

- **Completeness:** For all $(x, w) \in \mathcal{R}$, proofs generated by **Prove** are accepted by **Verify**,

$$\Pr[\text{Verify}(\text{crs}, x, \text{Prove}(\text{crs}, x, w)) = 1] = 1$$

- **Soundness:** No polynomial time adversary can produce a valid proof for a false statement,

$$\Pr[\text{Verify}(\text{crs}, x, \pi) = 1 \text{ for some } x \notin L_{\mathcal{R}}] \leq \text{negl}(\lambda)$$

- **Zero Knowledge:** There exists a simulator S that can generate proofs indistinguishable from real ones without knowledge of a witness¹,

$$\{\text{crs}, \text{Prove}(\text{crs}, x, w)\} \approx_c \{\text{crs}, S(\text{td}_{\text{sim}}, x)\}.$$

While NIZK proofs allow one to prove statements about NP relations without revealing witnesses, they do not by themselves provide a mechanism for binding a prover to specific hidden values across multiple proofs or computations. In many cryptographic constructions, including HS, it is essential to ensure that a prover cannot adaptively change its underlying data while still producing valid proofs. Commitment schemes address this limitation by enabling a party to bind to a value while keeping it hidden.

Definition 4 (Commitment). A commitment scheme consists of probabilistic polynomial time algorithms

1. $(c, r) \leftarrow \text{Commit}(m)$ where m is a message, c is the commitment and r is the opening randomness.
2. $\{0, 1\} \leftarrow \text{Open}(c, m, r)$ outputs whether the commitment correctly opens to the message m .

A commitment scheme must satisfy two important properties:

- **Hiding:** For any PPT adversary, commitments to different messages are computationally indistinguishable.

$$\text{Commit}(m_1) \approx_c \text{Commit}(m_2)$$

- **Binding:** It is computationally infeasible to find $m' \neq m$ and r', r such that

$$\text{Open}(c, m, r) = \text{Open}(c, m', r') = 1.$$

Commitment schemes provide a foundational mechanism for binding hidden data, but they do not support reasoning about computations performed on committed values. In applications such as HS and verifiable computation, one must verify not only that a value was committed correctly, but also that a publicly claimed output corresponds to a prescribed function applied to the committed input. Functional commitment schemes extend standard commitments with exactly this capability, enabling verifiable evaluation of functions over committed data.

Definition 5 (Functional Commitments). A functional commitment (FC) scheme allows committing to a message m while later enabling the verification of the value of a function $f(m)$ without revealing m . It comprises of the following algorithms:

- $\text{params} \leftarrow \text{FC.Setup}(1^\lambda)$, generates the public parameters.
- $(c, r) \leftarrow \text{FC.Commit}(\text{params}, m)$, outputs a succinct commitment c and an opening hint r on the input message m .
- $(y, \pi) \leftarrow \text{FC.Open}(\text{params}, r, f, x)$, given a function f and an input x , it computes the evaluation $y = f(x)$, and a succinct proof π that y is a correct evaluation of the committed value.
- $\{0, 1\} \leftarrow \text{FC.Verify}(\text{params}, c, f, y, \pi)$, takes in the commitment, the proof and verifies that $y = f(x)$ via the proof π for c .

With these algebraic and cryptographic foundations in place, we now review concrete homomorphic signature constructions, organized according to their supported function classes, i.e., linear or non-linear.

¹Here $\approx_c$ represents computational indistinguishability

2.2 Homomorphic Signatures basic construction

The work by Johnson et al. [JMSW02] is the first one which provides a rigorous definition of what a homomorphic signature is. The specifications needed to define a homomorphic signature scheme are the same as for the digital counterpart: the message space $\mathcal{M}$ equipped with an operation, the space of signatures $\mathcal{Y}$ (which in this case is also equipped with a homomorphic operation) and the space for the secret ($\mathcal{K}$) and the public ($\mathcal{K}'$) keys respectively. Of course, the key-generation, the signing, and the verifying algorithms are still absolutely necessary.

However, unlike standard digital signatures that authenticate isolated messages, a homomorphic signature scheme authenticates messages within the context of a collection of messages. This structure ensures that homomorphic operations are only performed on messages belonging to the same authenticated batch, typically linked by a unique identifier. More formally this collection of messages is referred to as a dataset.

Definition 6 (Dataset). A dataset is a finite, indexed collection of messages together with their authenticators, which serves as the domain of admissible homomorphic evaluation. More formally, given a message space $\mathcal{M}$ and let $\mathcal{T}$ be a space of unique identifiers. A dataset is a finite indexed family

$$\mathcal{D} = (\tau, I, \{m_i\}_{i \in I})$$

where $I \subseteq \mathbb{N}$ is a finite index set and $m_i \in \mathcal{M}$. The $\tau \in \mathcal{T}$ identifies the collection.

Now that we have defined the dataset, we proceed to define a Homomorphic Signature that is defined over a dataset. Formally, we have

Definition 7 (Homomorphic Signature). A homomorphic signature scheme HS is a tuple of the following PPT algorithms:

- **HS.Keygen**($1^\lambda, N$). Takes as input the security parameter λ and an integer N . It outputs a secret key $\mathbf{sk}$ and its respective public key $\mathbf{pk}$. The public keys determines the space of the messages $\mathcal{M}$, the space of the signatures $\mathcal{Y}$ and the set of admissible functions.
- **HS.Sign**($\mathbf{sk}, \tau, m, i$). Takes as input the secret key $\mathbf{sk}$, a dataset and its associated tag $\tau \in \{0, 1\}^\lambda$, a message $m \in \mathcal{M}$ and an index $i \in \{1, \dots, N\}$. It outputs a signature $\sigma \in \mathcal{Y}$, computed using the secret key, which is the signature for the i -th message m of the dataset tagged by τ .
- **HS.Eval**($\mathbf{pk}, \tau, \sigma, f_{\tau, \mathbf{pk}}$). It takes input the public key $\mathbf{pk}$, a tag $\tau \in \{0, 1\}^\lambda$, an admissible function $f_{\tau, \mathbf{pk}} \in F$ and a tuple of signatures $\sigma \in \mathcal{Y}^N$. It outputs a signature $\sigma' \in \mathcal{Y}$.
- **HS.Ver**($\mathbf{pk}, \tau, m, \sigma, f_{\tau, \mathbf{pk}}$). It takes as input a public key $\mathbf{pk}$, a tag $\tau \in \{0, 1\}^\lambda$, a message $m \in \mathcal{M}$, a signature $\sigma \in \mathcal{Y}$ and a function $f \in F$. It outputs ‘ok’ if σ is a valid signature for the message m , output for the function f over the dataset tagged by τ , under the public key $\mathbf{pk}$ (depicted by $f_{\tau, \mathbf{pk}}$).

2.3 Properties : Correctness, Succinctness, Security and Privacy

The utility of a Homomorphic Signature scheme is governed by a multi-dimensional trade-off between its functional capacity and its formal guarantees. In this section, we systematize the four fundamental pillars that characterize the HS landscape, Correctness, Succinctness, Security, and Privacy.

2.3.1 Correctness

In homomorphic signatures schemes, correctness requires that any signature produced, either directly by the signer (fresh signatures) or via homomorphic evaluation of honestly generated signatures must verify successfully for the corresponding message. In this systematization, we identify three paradigms of correctness. We distinguish between *absolute correctness* of pairing based schemes, *bounded correctness* of lattice based construction and *statement based correctness* of proof based frameworks.

In pairing based constructions, signatures are typically group elements in a bilinear setting $(\mathbb{G}_1, \mathbb{G}_2, \mathbb{G}_T)$. The homomorphic evaluation performs exact group operations. Verification succeeds if a pairing equation is satisfied. Since these operations are performed over finite fields, there is no noise growth. The evaluation is an exact algebraic mapping from $\mathbb{G}^N \mapsto \mathbb{G}$. Hence, the correctness is absolute.

Lattice based signatures, which rely on the SIS problem, exhibits a different correctness behaviour. In these construction, signatures are short integer vectors $\sigma \in \mathbb{Z}^n$ which satisfies a relation of the form

$$\mathbf{A} \cdot \sigma = u(m)$$

where $u(m)$ is some function over the message m and $\|\sigma\| \leq B$ where B is a verification bound derived from the SIS parameters. However, unlike group based elements, lattice signatures exhibit a growth in the norm. Every homomorphic operation increases the magnitude of the resulting signature. For an admissible function f , the norm of the evaluated signature σ^* is bounded by

$$\|\sigma^*\| \leq \mathcal{L}_f \cdot B$$

The growth factor $\mathcal{L}_f$ is determined by the structure of the homomorphic circuit.

- Linear schemes : Where $f(m_1, \dots, m_n) = \sum_{i=1}^n a_i m_i$, then

$$\mathcal{L}_f = \sqrt{\sum_{i=1}^n |a_i|^2}$$

- Polynomial schemes : For schemes satisfying polynomial functions of depth d ,

$$\mathcal{L}_f \leq (\text{poly}(n, \log q))^d$$

In the case of multi-key setup, the growth factor multiplies with a polynomial factor of the number of singers. We also note that there exists a correctness margin which could be defined by $B_{\text{ver}} - \mathcal{L}_f \cdot B$ where B_{ver} is the correctness verification bound. As this margin approaches zero, the schemes exhibits verification failure, and thus is one of the primary reasons why lattice based HS are often levelled.

2.3.2 Succinctness

One of the fundamental requirement of homomorphic signatures is succinctness. This dictates that the size of the evaluated signature must remain independent of the number of input messages. Unlike trivial authentication methods where signatures accumulate linearly with the dataset, homomorphic evaluation maps the different signatures into a compact signature. More formally it is defined as

Definition 8 (Succinctness [CFN18]). A homomorphic signature scheme is said to be succinct if there exists a polynomial $p(\lambda)$ such that for all $(\text{pk}, \text{sk}) \leftarrow \text{HS.Keygen}(1^\lambda, N)$, all $(m_1, \dots, m_N) \in \mathcal{M}^N$, and tag τ and all admissible functions $f_{\tau, \text{pk}}$, if $\sigma_i \leftarrow \text{HS.Sign}(\text{sk}, \tau, m_i, i)$ and $\sigma \leftarrow \text{HS.Eval}(\text{pk}, \tau, \sigma_1, \dots, \sigma_N, f_{\tau, \text{pk}})$, then $|\sigma| \leq p(\lambda) \cdot \log N$.

Succinctness is essential for scalability in application such as verifiable data aggregation or outsourced computation, where large datasets are reduced to compact authenticated outputs. However, succinctness is not a single faceted property. It is rather a multi-constrained : *data succinctness*, where the signature size is invariant to the dataset size, *key-succinctness*, where the signature remains constant regardless of the number of distinct public keys in a multi-key setting and finally *circuit-independent succinctness*, where the signature size does not expand with the depth of the computation. For the case of homomorphic signatures, data-succinctness is a fundamental and non-negotiable property. However, while schemes naturally combine multiple signatures into a compact output, they often struggle to maintain key-succinctness or circuit-independence. In multi-key setting, the signature must be able to show the contribution of each independent public key to prevent any kind of insider corruption. This often requires the signature to carry some kind of key-identifier, thus scaling the with the number of signers. Achieving key-succinctness thus requires heavier primitives like NIZKs. Similarly circuit-independence is not natural as the depth of the circuit increases, the proof of correct computation often grows. Therefore most signature would require additional elements to prove the final result is correctly derived from the source.

On the other hand, the mechanism of succinctness differs fundamentally on the underlying hard problem of the scheme. Perfect succinctness is standard in pairing based linear schemes. However, it becomes difficult to maintain once we move beyond linear or low-degree functional evaluation. In lattice based setting, succinctness is often bounded by the noise. Lattice signature are naturally single vector, the values in that vector naturally grows in magnitude as homomorphic operations are performed. So even if there is no growth on the number of elements, the bit size required to store it increases. Hence it does depend on the circuit, thus not achieving any circuit-independent succinctness.

2.3.3 Security / Unforgeability

We begin by noting that classical existential unforgeability under adaptive chosen-message attacks (EUF-CMA) is the standard security notion for digital signatures. Informally, it requires that no probabilistic polynomial-time adversary, even after adaptively obtaining signatures on messages of its choice, can produce a valid signature on a new message. However, we observe that this notion is fundamentally incompatible with HS.

Because HS are intentionally malleable, given valid signatures on a set of authenticated messages, anyone can publicly derive a valid signature on the result of an admissible computation over these messages. Such derived signatures would be considered forgeries under classical EUF-CMA, despite being legitimate outputs of a homomorphic signature scheme. Consequently, we move away from the classical form, and instead adopt *function-aware* unforgeability notions that explicitly account for homomorphic evaluation.

A distinctive feature of HS, compared to classical signatures, is that the notion of a forgery depends not only on the final message and signature, but also on the dataset and the function used to derive the message. As a result, the definition of unforgeability critically depends on *when* the adversary commits to the target of its forgery.

In *selective* security notions, the adversary must commit in advance to certain parameters of the target forgery, such as the dataset identifier or the admissible function to be evaluated, before receiving any signatures. In contrast, under *adaptive* security notions, the adversary may choose these parameters dynamically based on the signatures it observes. As shown by Catalano et al. [CFN18], adaptive homomorphic unforgeability is strictly stronger than its selective counterpart.

We argue that these distinctions are not merely a matter of adversarial power, but lead to genuinely different notions of unforgeability, as they determine which outputs are considered legitimate homomorphic evaluations and which constitute forgeries.

Definition 9 (Forgery [Fre12]). A forgery is defined as a valid signature on previously unseen message. More formally, a (m^*, σ^*) is said to be a forgery if $\text{Verify}(k, m^*, \sigma^*) = 1$ and $m^* \neq m_i \in \mathcal{M}$.

Definition 10 (Homomorphic Unforgeability [CFN18]). A homomorphic signature scheme is unforgeable if for all $N \in \mathcal{N}$ and any probabilistic polynomial-time adversary $\mathcal{A}$, in the following game, denoted by $\text{Exp}_{\mathcal{A}, \text{HS}}^{\text{EUF}}(\lambda)$ the advantage of $\mathcal{A}$ is negligible with respect to λ :

- **Setup:** Challenger generates the key-pair $(\text{sk}, \text{pk}) \leftarrow \text{HS.Keygen}(1^\lambda, N)$. The public-key k determines the space of messages $\mathcal{M}$, the space of signatures $\mathcal{Y}$ and the set of admissible functions $f : \mathcal{M}^N \mapsto \mathcal{M}$. The public-key is given to the adversary $\mathcal{A}$.
- **Sign Query:** Adversary $\mathcal{A}$ adaptively selects a dataset $\mathcal{D} = \{m_1, \dots, m_n\} \in \mathcal{M}^N$ and sends a query of the form $(\tau_{\mathcal{D}}, i, m_i)$. For each query, the challenger checks if the same tag, message, index triple has been queried before or not. If no, it returns the $(\tau_i, m_i, \sigma_i = \text{HS.Sign}(k, \tau_i, m_i, j))$, otherwise it returns the same signature to $\mathcal{A}$.
- **Forgery Output:** $\mathcal{A}$ outputs a tuple $(\tau^*, m^*, \sigma^*) \in \{0, 1\}^\lambda \times \mathcal{M} \times \mathcal{Y}$. The experiment outputs 1 if the tuple returned by $\mathcal{A}$ is a forgery, and 0 otherwise.

Completing the definition above, we now provide two further security definitions which actually define when the tuple returned is considered a forgery. Each of the two gives a security notion for the homomorphic signature scheme.

Definition 11 (Selective Security). We define this notion as a security game $\text{Exp}_{\mathcal{A}, \text{HS}}^{\text{sel-EUF}}(\lambda)$ which proceeds as $\text{Exp}_{\mathcal{A}, \text{HS}}^{\text{EUF}}(\lambda)$ with the addition condition that for the tuple returned by the adversary (τ^*, m^*, σ^*) , $\text{HS.Ver}(\text{pk}, \tau^*, m^*, \sigma^*, f_{\tau^*, \text{pk}})$ returns ‘ok’ and one of the following holds

- (Forgery I) the dataset denoted by τ^* has not been queried before;
- (Forgery II) for some dataset τ^* , the message is not a functional evaluation over queried messages, i.e., $m^* \neq f_{\tau^*, \text{pk}}(m_1, \dots, m_N)$.

Let $\text{Adv}_{\mathcal{A}, \text{HS}}^{\text{sel-EUF}}(\lambda) = \Pr[\text{Exp}_{\mathcal{A}, \text{HS}}^{\text{sel-EUF}}(\lambda) = \text{ok}]$ be the advantage of $\mathcal{A}$ against the selective security of the scheme HS. We say the homomorphic signature scheme is selectively secure if for every PPT adversary $\mathcal{A}$ there exists some negligible function $\epsilon(\lambda)$ such that $\text{Adv}_{\mathcal{A}, \text{HS}}^{\text{sel-EUF}}(\lambda) \leq \epsilon(\lambda)$.

Selective unforgeability is well suited to applications in which datasets are authenticated in a single batch, as in early homomorphic signature constructions, or in settings where homomorphic computations are performed only after the dataset has been completely populated. However, this notion provides weaker guarantees in dynamic environments, such as streaming applications, where datasets are signed incrementally and computations may be carried out on partially authenticated data. In such settings, an adversary may exploit the delayed commitment to dataset identifiers and produce a forgery associated with a dataset label that was not previously queried during the signing phase. This limitation motivates stronger notions of unforgeability that better capture dynamic and incremental computation scenarios.

Definition 12 (Adaptive Security). We define this notion as a security game $\text{Exp}_{\mathcal{A}, \text{HS}}^{\text{ad-EUF}}(\lambda)$ which proceeds as $\text{Exp}_{\mathcal{A}, \text{HS}}^{\text{EUF}}(\lambda)$ with the additional condition that for the tuple returned by the adversary (τ^*, m^*, σ^*) , $\text{HS.Ver}(\text{pk}, \tau^*, m^*, \sigma^*, f_{\tau^*, \text{pk}})$ returns ok and one of the following holds

- (Forgery I) the dataset denoted by τ^* has not been queried before;

- (Forgery II) for some dataset τ^* , the message is not a functional evaluation over queried messages, i.e., $m^* \neq f_{\tau^*, \text{pk}}(m'_1, \dots, m'_N)$, where $m'_i = m_i$ for all queries messages, and $m'_i = \tilde{m}$ (some arbitrary $\tilde{m} \in \mathcal{M}$) otherwise.
- (Forgery III) the functions f is not well defined with respect to the messages queried.

Let $\text{Adv}_{\mathcal{A}, \text{HS}}^{\text{ad-EUF}}(\lambda) = \Pr[\text{Exp}_{\mathcal{A}, \text{HS}}^{\text{ad-EUF}}(\lambda) = \text{ok}]$ be the advantage of $\mathcal{A}$ against the adaptive security of the scheme HS. We say the homomorphic signature scheme is adaptively secure if for every PPT adversary $\mathcal{A}$ there exists some negligible function $\epsilon(\lambda)$ such that $\text{Adv}_{\mathcal{A}, \text{HS}}^{\text{ad-EUF}}(\lambda) \leq \epsilon(\lambda)$.

For a general homomorphic signature scheme, it may not be possible for the challenger to efficiently detect a Forgery II (i.e., determine whether the function f^* output by the adversary is well-defined on F). However, for linearly homomorphic schemes or cases where the admissible function are low-degree arithmetic circuits over a large field, it is easy to detect such a forgery. As a result, we can convert an adversary that outputs a Forgery III into one that outputs a Forgery II [Fre12, CFN18].

A general technique for upgrading selective unforgeability to fully adaptive unforgeability was introduced in [GVW15]. Their approach relies on combining selectively secure homomorphic signature schemes with additional cryptographic tools that allow binding of dataset identifiers. At a high level, the transformation leverages homomorphic trapdoor functions (see Section 3.3.1) to encode dataset-dependent information into the signing process, together with chameleon hash functions to enable equivocation of dataset identifiers in the security reduction. This allows the simulator to defer fixing the target dataset until after observing the adversary's queries, thereby overcoming the main obstacle in proving adaptive security.

Theorem 3 (Selective to Adaptive Security [GVW15, CFN18]). *Assuming the existence of homomorphic trapdoor functions and chameleon hash functions, any selectively unforgeable homomorphic signature scheme for a bounded class of functions can be transformed into a scheme that is secure against adaptive adversaries. The transformation incurs a polynomial overhead in the security parameter and the supported circuit depth.*

2.3.4 Privacy / Context Hiding

Homomorphic signatures are sometimes needed to satisfy another property called *context hiding* [BF11b]. The main idea underlines that signatures on function's output must not reveal information about the original signed inputs. More informally, given signatures $\sigma_1, \dots, \sigma_n$ over inputs $m_1, \dots, m_n$, the signature σ' on a derived message $m' \in \text{span}(m_1, \dots, m_n)$ must not leak any information about $m_1, \dots, m_n$ beyond what is revealed by m' . A more formal definition follows.

Definition 13 (Weakly Context Hiding). [ABC⁺12] A homomorphic signature scheme is said to be weakly context hiding if σ' only reveals information about the corresponding message m' , but does not leak any information about the dataset $m_1, \dots, m_n$ of the corresponding signature $\sigma_1, \dots, \sigma_N$. However, it might leak some partial structural information about the inputs.

Ahn et al. define a stronger notion that requires that the derived signatures be distributed as independent fresh signatures on the same message. This ensures privacy even if the original signatures $\sigma_1, \dots, \sigma_n$ are exposed.

Definition 14 (Strong Context Hiding). [ABC⁺12] A homomorphic signature scheme is said to be strong context hiding if σ' , that has been derived from $\sigma_1, \dots, \sigma_N$, is statistically indistinguishable from any fresh signature on the message m' from the dataset M . Additionally, even if two set of messages map to the same message m' via the

function f , the signatures derived on the two datasets are statistically and computationally indistinguishable.

The difference with weak context hiding is that the earlier only ensures privacy when the original signatures remain hidden. Additionally, as highlighted in [ALP12], the notion of strong context hiding, takes only the indistinguishability from honestly generated signatures into account. This implies that even if the attacker generated the signature using an admissible function [AL11a], this passes the case. To address this linkability with signatures chosen by attacker, authors of [ALP12] proposed a stronger notion.

Definition 15 (Complete Context Hiding). [ALP12] A homomorphic signature scheme is said to be completely context hiding if some σ^* , that has been generated dishonestly (just using f and m' but not the original message set $m_1, \dots, m_N$), is statistically and computationally indistinguishable from a honestly derived signature σ' using HS.Eval over f and m' .

2.4 Security models : Random Oracle Model vs Standard model

The security of homomorphic signature schemes is typically established either in the Random Oracle Model (ROM) or in the standard model, and this distinction has important implications for both unforgeability and privacy guarantees, particularly context hiding.

The Random Oracle Model idealizes hash functions as truly random functions that are globally accessible to all parties, including the adversary. Proofs in the ROM may program the oracle adaptively, allowing the simulator to embed challenges or extract information in ways that are not known to be realizable with any concrete hash function. As a result, security in the ROM should be interpreted as heuristic: while ROM-based schemes are often efficient and conceptually simpler, their security relies on an idealization that may not hold in practice.

The advent of lattice-based HS necessitates a transition from the classical ROM to the Quantum Random Oracle Model (QROM) [BDF⁺11]. While the ROM assumes adversaries make discrete, classical queries to a hash function, the QROM accounts for quantum-capable adversaries who can evaluate the oracle on a superposition of inputs. This shift introduces significant technical hurdles: many classical proof techniques, such as adaptive programming [BDF⁺11] and rewinding [Wat06, Unr12], do not trivially hold in the QROM due to the no-cloning theorem and the collapse of the quantum state upon measurement. For lattice-based HS, this means that simple Fiat-Shamir transformations, which are often used to achieve context hiding or non-interactivity, require more sophisticated constructions [Unr15] or lossy identification [AFLT16] schemes to maintain unforgeability. Consequently, while lattice-based schemes are inherently resistant to Shor’s algorithm, their security reductions in the QROM are often more complex and may result in a loss of tightness compared to their classical ROM counterparts [KLS18].

In contrast, proofs in the standard model (SM) rely only on well-defined cryptographic assumptions, such as hardness of lattice problems or bilinear group assumptions, without idealized oracles. Security guarantees obtained in the standard model are therefore stronger and more robust, albeit typically at the cost of increased complexity or larger signatures.

The distinction between ROM and standard model is even more pronounced when considering context hiding. Weak context hiding is often achievable in the ROM by hashing messages or randomness into the oracle to mask auxiliary data [BF11a]. On the other hand, achieving strong or complete context hiding in the standard model typically requires structured commitments [GVW15] or algebraic proofs [GS08]. These tools ensure that derived signatures reveal no information beyond the final output, providing what we consider to be more interpretable and long-term privacy guarantees.

Ultimately, the choice between models reflects a strategic trade-off. We argue that while ROM-based schemes are attractive for practical, efficient deployment, standard-model

constructions remain the gold standard for high-security settings where reliance on idealized assumptions is undesirable.

Feature	ROM	Standard Model
Security type	Heuristic/Idealised	Reductionist / Concrete Assumptions
Efficiency	Compact Signatures	Heavier Proof systems
Context Hiding	Often weak	Strong/Statistically hiding
Common tools	Fiat-Shamir, Hash	NIZKs, Lattice trapdoors

3 Single-Key Homomorphic Signatures

We begin the survey by examining the single-key setting, which represents the most widely studied architecture in homomorphic signature landscape. In this, a single data owner (the signer) authenticates a dataset and an evaluator performs computation on that dataset. We categorize the single-key schemes into broadly three parts: firstly linear constructions, which permit addition, secondly, non-linear and polynomial constructions and finally the schemes that support arbitrary circuits.

3.1 Linear Homomorphic Signatures

Early constructions focused on supporting linear computations, particularly summations and scalar multiplications, over authenticated data. Formally, a linear homomorphic signature scheme allows one to compute a valid signature on a message $m' = \sum_{i=1}^n a_i m_i$ by combining signatures σ_i on m_i for $i \in [N]$ and coefficients a_i from a message space $\mathcal{M}$. The scheme consists of the following polynomial-time algorithms:

- $\text{Keygen}(1^\lambda) \rightarrow (\text{pk}, \text{sk})$: Generates a public (pk) and private key pair (sk).
- $\text{Sign}(\text{sk}, \tau, m, i) \rightarrow \sigma_i$: Generates a signature on message m_i from a dataset tagged by τ .
- $\text{Eval}(\text{pk}, \sigma_1, \dots, \sigma_n, (a_1, \dots, a_n)) \rightarrow \sigma'$: Produces a signature σ' on the linear combination $m' = \sum_i a_i m_i$.
- $\text{Verify}(\text{pk}, m', \sigma') \rightarrow \{0, 1\}$: Verifies whether σ' is a valid signature for m' .

The very first Homomorphic Signatures schemes introduced were linear. Even though Boneh et al. [BFKW09] is widely regarded as the first big milestone in literature, however, many schemes such as [CJL09, ZKMH07, YWRG08, YCK10] have been proposed, but they have been already proved to be impractical and not completely secure [Wan10, Zha10, DCNR11].

In the last decade and half, a string of new constructions have come to fore, which have been shown to be secure under either the Random Oracle model or the Standard security model.

3.1.1 Boneh-Freeman-Katz-Waters Scheme [BFKW09]

The paper "Signing a Linear Subspace: Signature Schemes for Network Coding" develops homomorphic signature schemes grounded in the algebraic structure of vector spaces. These constructions allow signers to produce compact signatures on a set of vectors forming a linear subspace, such that anyone can publicly verify the authenticity of any linear combination within that subspace without accessing the original signatures. Mathematically, the schemes leverage bilinear pairings and dual orthonormal bases to achieve signature

homomorphism over finite fields, facilitating efficient verification of linear relations among signed data. The first scheme achieves strong homomorphic properties under the random oracle model with constant-size public parameters, while the second offers comparable guarantees under standard complexity assumptions like the Decisional Linear (DLIN) and Strong Diffie-Hellman (SDH) problems. The scheme offers a lower bound the signatures length giving an optimal construction. Although the scheme was only investigated in the context of a weak adversary, the signatures enjoy strong context hiding [ALP13].

This scheme supports signing of vectors in $\mathbb{F}_p^n$ and allows public derivation of signatures on any linear combination of previously signed vectors. It is constructed in a bilinear group setting.

- **Keygen:** Let $\mathcal{G} = (\mathbb{G}_1, \mathbb{G}_2, \mathbb{G}_T, p)$ be a bilinear group tuple of prime order p with generator $g \xleftarrow{\$} \mathbb{G}_2 \setminus \{1\}$. Choose secret key $\alpha \in \mathbb{F}_p$, an hash function H and set $u = g^\alpha$. Set public key $\mathbf{pk} = (\mathcal{G}, H, g, u)$ and the secret key $\mathbf{sk} = \alpha$.
- **Sign:** Given the secret key $\mathbf{sk}$, and a message vector $\mathbf{m} = (m_1, \dots, m_n) \in \mathbb{F}_p^n$ and a dataset identifier τ , the signature σ is computed as

$$\sigma = \left(\prod_{i=1}^n H(\text{id}, i)^{m_i} \right)^\alpha$$

- **Homomorphic Evaluation:** Given signatures $\sigma_1, \dots, \sigma_k$ and a function f represented by its coefficients $a_1, \dots, a_k \in \mathbb{F}_p$, compute a signature σ on $m = \sum_{j=1}^k a_j m_j$ as:

$$\sigma = \prod_{j=1}^k \sigma_j^{a_j}$$

- **Verify:** Given message vector $\mathbf{y} \in \mathbb{F}_p^n$, the identifier and signature σ , accept if:

$$e(\sigma, g) \stackrel{?}{=} e \left(\prod_{i=1}^n H(\text{id}, i)^{y_i}, u \right)$$

3.1.2 Gennaro-Katz-Krawczyk-Rabin Scheme [GKKR10]

This work proposed the first homomorphic signature scheme based solely on the RSA assumption (in the random oracle model), and a homomorphic hash function based on the hardness of factoring. Unlike prior schemes that rely on bilinear maps and large finite fields, this work proposes operating directly over the small integers, enabling efficient computation and reduced bandwidth overhead. The scheme has been proven unforgeable only against weak adversary.

Here is the summary of the scheme.

- **Keygen:** Choose a hash function $H : \{0, 1\}^* \rightarrow \mathbb{Z}_N$ for a large RSA modulus N . The signer selects random elements (generators) $g_1, \dots, g_n \in \mathbb{Z}_N^*$ of the subgroup of quadratic residues. The public key is $(N, e, g_1, \dots, g_n)$ while the private key is d such that $ed \equiv 1 \pmod{\phi(N)}$.
- **Sign:** To sign a message vector $\mathbf{v} = (v_1, \dots, v_n) \in \mathbb{Z}^n$:
 1. Compute the signature as:

$$\sigma = \left(\prod_{i=1}^n g_i^{v_i} \right)^d \pmod{N}$$

2. Output $\sigma \in \mathbb{Z}_N^*$

- **Homomorphic Evaluation:** Given signatures $\sigma_1, \dots, \sigma_k$ on vectors $\mathbf{m}_1, \dots, \mathbf{m}_k$, and integer coefficients $a_1, \dots, a_k$, compute a signature on the linear combination $\mathbf{m} = \sum_{j=1}^k a_j \mathbf{m}_j$ as:

$$\sigma = \prod_{j=1}^k \sigma_j^{a_j} \pmod{N}$$

The result σ is a valid signature on $\mathbf{m}$. Note again like the last scheme the signatures are linear in the exponent.

- **Verify:** Given $\mathbf{y}$ and signature σ , accept if:

$$\sigma^e \stackrel{?}{=} \left(\prod_{i=1}^n g_i^{y_i} \right) \pmod{N}$$

3.1.3 Boneh-Freeman Scheme Over Binary fields [BF11b]

This paper proposed the first work to authenticate vectors over binary fields. The scheme is based on the hardness assumption of the SIS problem, to which reductions were established from the standard intractable lattice problems SIVP and GapSVP [MR07]. This scheme extends the ideas of that presented in [GPV08] where the signatures are short vectors in lattices.

- **Keygen:** Let H be a hash function. The signer generates a matrix and trapdoor pair $\mathbf{A}, T_A \leftarrow \text{TrapGen}(2q, n)$ and publishes the public key $\text{pk} = (\mathbf{A}, H)$. The secret key sk is the trapdoor basis T_A for the lattice $\Lambda_{2q}^\perp(\mathbf{A})$.
- **Sign:** Given a message vector $m \in \mathbb{F}_2$ and an identifier $\tau \in \{0, 1\}^t$, the signer samples a vector σ from the lattice $\Lambda_{2q}^\perp(\mathbf{A} \| H(\tau))$, using

$$\sigma \leftarrow \text{SamplePre}(\mathbf{B}, T_B, q \cdot m, \delta)$$

where $\mathbf{B} = (\mathbf{A} \| H(\tau))$ and T_B is the basis for $\mathbf{B}$ such that $\mathbf{B} \cdot \sigma = q \cdot m \pmod{2q}$.

- **Homomorphic Evaluation:** Given signatures $\sigma_1, \dots, \sigma_k$ on messages $m_1, \dots, m_k$ and constants $\alpha_i \in \mathbb{F}_2, i \in [k]$, one can derive a signature on the linear combination of these messages:

$$\sigma = \sum_{i=1}^k \alpha_i \sigma_i$$

The resulting σ is a valid signature on $m = \sum_i \alpha_i m_i$, as both linearity and shortness are preserved.

- **Verify:** Given the public key $(\mathbf{A}, H)$, an identifier τ , message y and signature σ , compute $\mathbf{B} = (\mathbf{A} \| H(\tau))$ and verify:

$$\mathbf{B} \cdot \sigma \stackrel{?}{=} q \cdot y \pmod{2q} \quad \text{and} \quad \|\sigma\| \leq \mathbf{L}$$

for some norm bound $\mathbf{L}$.

Remark 1. Another paper, namely "Lattice-based Linearly homomorphic signature scheme over binary fields" by Wang, Hu and Wang [WHW13] was proposed as a sequel to [BF11b] and relies on the hardness of the SIS problem. However, no improvements were reported in terms of security and privacy.

Remark 2. As observed in [Jin14], lattice-based homomorphic signature schemes over binary fields, such as the [BF11b], are inherently *L-limited*. That is, they only support at most L homomorphic combinations of signed messages, where L is tied to the structural parameters of the derived SIS instance. To date, despite advances including [WHW13] and more recently tight security lattice-based constructions in the standard model [GTLZ24], no scheme has overcome this limitation. In particular, there remains no candidate that provides both *unbounded homomorphic combination capacity* and *post-quantum lattice-based security* over $\mathbb{F}_2$.

3.1.4 Boneh-Freeman Scheme for Linear functions over Small Fields [BF11a]

This scheme was proposed as generalization over the previous scheme while extending it beyond $\mathbb{F}_2$ and providing verification of functional evaluation. It is based on the SIS problem on q -ary lattices for some prime q and takes inspiration from the hash-and-sign GPV scheme [GPV08]. The system is secure only for small p , with $p = \text{poly}(n)$ and $p \leq \sqrt{q/nk}$ for data set of size k . This scheme performs a authentication of both the messages and the function and binds them together.

- **Keygen:** Let p, q be two primes with $q \geq (nkp)^2$ (k is the dataset size) and also $\ell = \lceil n/6 \log q \rceil$. Define a lattice $\Lambda_1 = p\mathbb{Z}^n$, generate $(\mathbf{A}, T_A) \leftarrow \text{TrapGen}(n, q)$ and set $\Lambda_2 = \Lambda_q^\perp(\mathbf{A})$. Define $T = p \cdot T_A$, where T is the basis of $\Lambda_1 \cap \Lambda_2$. Like the previous scheme, consider a hash function $H : \mathbb{Z}^k \mapsto \mathbb{F}_q^\ell$. To bind a function f defined by a vector of coefficients $(c_1, \dots, c_n)$, define another hash function $\omega_\tau(\langle f \rangle) = \sum_i c_i H(\tau || i)$. Additionally, there are two isomorphism map from $\Phi_1 : \mathbf{x} \mapsto (\mathbf{x} \bmod \Lambda_1)$ and $\Phi_2 : \mathbf{x} \mapsto (\mathbf{x} \bmod \Lambda_2) = \mathbf{A}\mathbf{x} \bmod q$. Finally, publish the public key $\text{pk} = (\Lambda_1, \Lambda_2, k, H)$ and the secret key is $\text{sk} = T$.
- **Sign:** Given a message vector $\mathbf{m} \in \mathbb{F}_p^n$, a tag $\tau \in \{0, 1\}^n$, an index i and the secret key sk , the signer computes $\alpha_i = H(\tau || i)$ and then computes a $\mathbf{t} \in \mathbb{Z}^n$ such that $\mathbf{t} \bmod p = \mathbf{m}$ and $\mathbf{A} \cdot \mathbf{t} \bmod q = \alpha_i$. Finally to produce the signature, the signer performs

$$\sigma \leftarrow \text{SamplePre}(\Lambda_1 \cap \Lambda_2, T, \mathbf{t}, \delta)$$

- **Homomorphic Evaluation** Given the public key, the tag τ , a function f and a tuple of signatures $\sigma_1, \dots, \sigma_k \in \mathbb{Z}^n$, output

$$\sigma = \sum_{i=1}^k c_i \sigma_i$$

- **Verify:** Given the public key, a tag τ , a message vector $\mathbf{y}$, a signature σ and a function f , if all the following hold,
 - σ is small
 - $\Phi_1(\sigma) = \sigma \bmod p \stackrel{?}{=} \mathbf{y}$, (Message Authentication)
 - $\Phi_2(\sigma) = \mathbf{A} \cdot \sigma \bmod q \stackrel{?}{=} \sum_{i=1}^k c_i \alpha_i \bmod q$, (Function binding).

3.1.5 Attrapadung-Libert-Peters Schemes [AL11a, ALP12, ALP13]

In PKC 2011, Attrapadung and Libert presented the first *homomorphic network coding signature* scheme supported by a security proof in the *standard model* [AL11a]. Prior constructions either required signing entire sub-spaces at once or relied on the random oracle heuristic, limiting their practicality and provable guarantees. The innovation lies in allowing the signer to produce signatures on individual vectors *on-the-fly*, with each

signature incurring a small, constant-size overhead per packet, and without using idealized hash functions.

Their construction adapts the *dual encryption technique* originally introduced in [Wat09], combining public-key encryption, like structure with bilinear group operations to support homomorphic evaluation. The public key includes group elements that encode vector coordinates, while the signing algorithm generates compact proofs whose homomorphic recombination mirrors packet processing at intermediate network nodes. Verification requires checking pairing equations over a small constant number of group elements. Here is a small summary of the scheme.

- **Keygen:** Let $\mathcal{G} = (\mathbb{G}, \mathbb{G}_T, p)$ be bilinear group tuple of prime order $N = p_1 p_2 p_3$. Choose with generators $g_1 \in \mathbb{G}_{p_1}$, $g_3 \in \mathbb{G}_{p_3}$. Compute the target group element $e(g_1, g_1)^\alpha$, and the public parameters $u = g_1^b$, $h_i = g_1^{a_i}$ for $i \in [0, 1]$. The secret key is $\text{sk} = (g_1^\alpha, \kappa)$ while the public key is $\text{pk} = (\mathcal{G}, g_1, u, h_1, \dots, h_n, g_3)$.
- **Sign:** To sign a message vector $\mathbf{m} = (m_1, \dots, m_n) \in \mathbb{F}_p^n$ with identifier τ , compute a randomness $r \in \Psi(\kappa, \tau)$ and generate a session specific randomness $\text{id}' = H(\tau, e(g, g)^r)$. The signature is a triple $\sigma = (\sigma_1, \sigma_2, \sigma_3)$ which are computed as

$$\sigma^{(1)} = g^\alpha \cdot (u \cdot h_0^{\text{id}'}) \cdot r_3, \quad \sigma^{(2)} = g^r \cdot r'_3, \quad \sigma^{(3)} = \left(\prod_{i=1}^n h_i^{m_i} \right)^r \cdot r''_3$$

where $r_3, r'_3, r''_3 \xleftarrow{\$} \mathbb{G}_{p_3}$.

- **Homomorphic Evaluation:** Given signatures σ_i , the public key pk and the coefficients of the linear combination $(\beta_1, \dots, \beta_n)$, the evaluator keeps the first two components of the signature, i.e. $\sigma^{(1)} = \sigma_1^{(1)} \cdot r_3$ and $\sigma^{(2)} = \sigma_1^{(2)} \cdot r'_3$. The third component is computed as follows $\sigma^{(3)} = \prod \sigma_i^{(3)\beta_i} \cdot r''_3$. The randomness (r_3, r'_3, r''_3) is picked fresh every time.
- **Verify:** Given a message $\mathbf{y}$, signature σ , the public key pk and a identifier τ , the verifier checks the two equations:

$$\begin{aligned} 1. \quad & e(\sigma^{(1)}, g) \stackrel{?}{=} e(g, g)^\alpha \cdot e(u \cdot h_0^{\text{id}'}, \sigma^{(2)}) \\ 2. \quad & e(\sigma^{(3)}, g) \stackrel{?}{=} e(\sigma^{(2)}, \prod_j h_j^{y_j}) \end{aligned}$$

The resulting scheme achieves full EUF-CMA security without relying on random oracles, and supports public verifiability with minimal communication overhead. It thus provides a practical and provably secure foundation for defending against pollution attacks in network coding.

While the previous construction successfully established a framework for linearly homomorphic signatures in the standard model, the resulting signatures were only weakly context-hiding. To address this privacy limitation, [ALP12] refined the composite-order group approach by formalizing the use of the G_{p_3} subgroup as a dedicated blinding layer. By transitioning from the three-tuple structure of previous scheme to a more compact two-tuple $\sigma = (\sigma^{(1)}, \sigma^{(2)})$, this work ensured that the G_{p_3} components effectively mask the linear combination of the G_{p_1} base elements. This mathematical pivot allowed the scheme to achieve statistical context hiding, where the final signature becomes indistinguishable from a fresh one produced by the signer, even when the evaluator lacks the master secret α .

Going in little detail, the construction of [ALP12] achieves complete context hiding by leveraging the orthogonality of composite order bilinear groups where $N = p_1 p_2 p_3$. In this setting, the public key includes generators g_1, u, v and message bases $\{h_i\}_{i=1}^n$ all sampled

from the $\mathbb{G}_{p_1}$ subgroup, while a generator g_3 provides access to the blinding $\mathbb{G}_{p_3}$ subgroup. A signature on a vector $\vec{v}$ is a two-tuple $\sigma = (\sigma_1, \sigma_2)$ where $\sigma_1 = (\prod h_i^{v_i})^\alpha \cdot (u^\tau v)^r \cdot R_3$ and $\sigma_2 = g_1^r \cdot R'_3$, with R_3, R'_3 acting as randomizing elements in G_{p_3} . Because the pairing $e(\mathbb{G}_{p_1}, \mathbb{G}_{p_3})$ results in the identity element in $\mathbb{G}_T$, these blinding factors vanish during verification—defined by the relationship $e(\sigma_1, g_1) = e(\prod h_i^{y_i}, g_1^\alpha) \cdot e(u^\tau v, \sigma_2)$ —ensuring that the evaluated signature is statistically indistinguishable from one generated directly by the secret key holder.

Their 2013 work [ALP13] further refines the CoAD primitive, achieving completely context-hiding, quotable, linearly HS. This means that the derived signature is indistinguishable from a freshly generated one on the same message.

In some further detail, the construction achieves the complete context hiding in prime-order bilinear groups by replacing raw signature components with Groth-Sahai (GS) commitments and Non-Interactive Witness Indistinguishable (NIWI) proofs. While a foundational weakly context-hiding scheme (presented in Appendix F of [ALP12]) exposes the scalar s used to link the signature to its linear subspace, this work masks the signature elements $\sigma_1, \sigma_3, \sigma_4$ within GS commitments C_{σ_i} . The authentication of the message vector is verified through two pairing equations (Eq. 9 and 10 in [ALP13]) that operate directly on the committed values and their associated proofs π_1, π_2 . The signature is a six tuple of elements $(C_{\sigma_1}, \sigma_2, C_{\sigma_3}, C_{\sigma_4}, \pi_1, \pi_2)$. A critical feature of this scheme is the re-randomizability of the GS proofs; during the homomorphic addition process, an evaluator can transform a combined signature into a statistically independent version. This ensures that the derived signature leaks no information about the coefficients of the function f used in the linear combination, providing a privacy-preserving chain of custody in the standard model.

Together, these works significantly advance the state of standard-model HS, introducing features such as quotability, strengthened privacy guarantees, and refined security models, all while retaining constant-size signatures and efficient public evaluation.

3.1.6 Catalano-Fiore-Warinschi schemes [CFW11, CFW12]

The work in [CFW11] defines a strong group-theoretic assumption called *adaptive pseudo-freeness*, which generalizes the notion of free groups². The scheme supports signing vectors and deriving signatures on their linear combinations, making it suitable for secure network coding. The authors instantiate this construction using RSA groups with strong hardness guarantees. This construction is specifically designed to work where a file is a vector space V spanned by vectors $(v^{(1)}, \dots, v^{(m)})$. To ensure security, the vectors $v^{(i)}$ are augmented with unitary vectors $u^{(i)}$ and a random mask s_i .

- **Keygen:** For a given group $\mathbb{G}$ of some order, generate some elements $A = (g, g_1, \dots, g_n, h_1, \dots, h_m)$. Let $X = \{x\}$ be a set of constant symbols. The public key is set to be $\text{pk} = (X, A, \mathbb{G}, e, \{s_i, u^{(i)}, v^{(i)}\}_{i=1}^m)$ and the secret key $\text{sk} = \text{order}(\mathbb{G})^3$, where $e = H(\tau)$.
- **Sign:** To sign a file with identifier τ , and its associated V , for the i th vector find the element σ'_i that satisfies the variable x_i in the following equation in the group

$$x_i^e = g^{s_i} \prod_{j=1}^m h_j^{u_j^{(i)}} \cdot \prod_{j=1}^n g_j^{v_j^{(i)}}$$

The signature for i th element is $\sigma_i = (e, s_i, u^{(i)}, v^{(i)}, \tau, \sigma'_i)$ and the final signature is $\sigma = (\sigma_1, \dots, \sigma_m)$ for V .

²Informally, a group is pseudo-free if it is computationally hard for an adversary to find non-trivial group equations that evaluate to the identity—beyond what's implied by group axioms—despite access to previously seen equations and solutions.[CFW11]

- **Homomorphic Evaluation:** To combine the signatures σ_i using random coefficients $\alpha_i \in \mathbb{Z}_q$, compute the new vector $u^{(i)} = \sum_{j=1}^n \alpha_j u_j^{(i)}$ and $v^{(i)} = \sum_{j=1}^n \alpha_j v_j^{(i)}$. The new signature value is computed as

$$\sigma' = \prod_{j=1}^n \sigma_j'^{\alpha_j}, \quad s = \sum_{i=1}^n \alpha_i s_i$$

- **Verify:** The verifier checks if the resultant (s, u, v, σ') satisfies the equation

$$\sigma'^e = g^s \prod_{j=1}^m h_j^{u_j} \cdot \prod_{j=1}^n g_j^{v_j}.$$

Building on their 2011 framework, the authors later presented a highly efficient instantiation in pairing-friendly groups [CFW12]. The construction avoids RSA-type groups and instead relies on the *q-Strong Diffie-Hellman (q-SDH)* assumption in bilinear groups. Here is a brief description of their scheme.

- **Keygen:** Let G_1, G_2 be bilinear groups of prime order p , with pairing $e : G_1 \times G_2 \rightarrow G_T$. Let $g \in G, g' \in G'$ be two generators. Choose a random $z \in \mathbb{Z}_p$. Like the previous scheme choose random elements $(h, h_1, \dots, h_m, g_1, \dots, g_n)$ from G . Publish the public key as $\text{pk} = (p, g, g', g'^z, h, h_1, \dots, h_m, g_1, \dots, g_n)$ while the secret key remains $\text{sk} = z$.
- **Sign:** To sign a augmented message vectors $(u, v) \in \mathbb{F}_q^{m+n}$, for some file identifier τ , pick some random mask $s \in \mathbb{Z}_p$ and compute

$$\sigma' = \left(h^s \prod_{j=1}^m h_j^{u_j} \cdot \prod_{j=1}^n g_j^{v_j} \right)^{\frac{1}{z+\tau}}$$

Output the signature $\sigma = (\sigma', s)$.

- **Homomorphic Evaluation:** The evaluation process follows the same steps as the previously described scheme i.e. one computes $\sigma' = \prod_{i=1}^n \sigma_i'^{\alpha_i}$ and $s = \sum_n \alpha_i \cdot s_i$.
- **Verify:** Upon receiving σ and message (u, v) , the verifier checks pairing-based relations using public parameters and verifies the following,

$$e(\sigma', g'^z \cdot g'^\tau) \stackrel{?}{=} e(h^s \prod_{j=1}^m h_j^{u_j} \cdot \prod_{j=1}^n g_j^{v_j}, g').$$

Verification remains efficient and publicly verifiable via pairing operations. The work eliminates the need for random oracles, achieves compact signatures, and retains the security guarantees of the adaptive pseudo-free group abstraction, delivering one of the most practical homomorphic signature schemes in the standard model.

3.1.7 Freeman : the general framework [Fre12]

Freeman, with his [Fre12] paper provided with a framework that transforms any *ordinary* signature scheme with two key properties into a linearly homomorphic signature scheme : (a) *linearity* - signatures embed message vectors so that linear combinations of signatures correspond to linear combinations of messages, (b) *tag-compatibility* - each data set is identified by a unique “tag” whose signature interacts correctly with message embedding. In particular, the work converted four schemes, [Wat05] (under the *CDH*), [BB04] (under

the q -SDH), [GHR99] (under *strong RSA*) and [HW09] (under *RSA*). Freeman preserved the underlying security model of the base signature schemes. Additionally, the scheme also chooses the use of a Hash function.

Definition 16. [Fre12] Let $\mathcal{S} = (\text{Keygen}, \text{Sign}, \text{Verify})$ be a signature scheme. Let $\mathcal{M}$ be the message space and Φ be the space of randomness sampled by the signing algorithm. We say the $\mathcal{S}$ is pre-homomorphic if the following three conditions hold for each key-pair (pk, sk) output by Keygen .

1. There is a finite cyclic group $\mathbb{G}$ such that Sign defines a map

$$\text{Sign}_{\text{sk}} : \mathcal{M} \times \Phi \mapsto \mathbb{G} \times \{0, 1\}^*,$$

We decompose the signature σ as (σ_1, σ_2) with $\sigma_1 \in \mathbb{G}$ and allow σ_2 to be free of choice.

2. the public key pk has a generator g of the group $\mathbb{G}$ and there is an efficiently computable function $f_{\text{sk}} : \mathcal{M} \times \Phi \mapsto \mathbb{Z}$ such that for each signature we have $\sigma_1 = g^{f_{\text{sk}}(m)}$

Note 1. [Fre12] Any signature scheme is pre-homomorphic just by taking the group $\mathbb{G} = 1$ and the function $f_{\text{sk}} = 0$ and let the σ_2 be the signature.

So given a pre-homomorphic signature $\mathcal{S}$, here is how it can be converted into a homomorphic signature.

- **Keygen:** Run $\mathcal{S}.\text{Keygen}$ to obtain $(\text{pk}_{\mathcal{S}}, \text{sk}_{\mathcal{S}})$. Let $\mathbb{G}, \mathbb{G}'$ be finite cyclic groups as declared in Definition 16 and R be a ring of exponents for $\mathbb{G}$. Let hk be a homomorphic hash key for the function H (details have been omitted, refer to [Fre12] for further details). Choose random $(t_1, \dots, t_k, u) \xleftarrow{\$} \mathbb{G}'$. Publish $\text{pk} = (\text{pk}_{\mathcal{S}}, hk, \{t_i\}_{i=1}^k, u, R)$ as the public key, and $\text{sk} = (\text{sk}_{\mathcal{S}}, \text{pk})$ as the secret key.
- **Sign:** To sign a message vector $\mathbf{m} = (m_1, \dots, m_n) \in R^n$ with tag τ , use the secret key sk to compute $(\sigma_1, \sigma_2) \leftarrow \mathcal{S}.\text{Sign}(\text{sk}_{\mathcal{S}}, \tau)$ binding the signature to the tag. Pick a randomness r , and derive a per-coordinate key $sk' = f_{\text{sk}}(\tau, r)$. Now for a choice of $i \in \{1, \dots, k\}$, compute $\sigma_3 = (t_i \cdot H(\mathbf{m}) \cdot u^r)^{f_{\text{sk}}(\tau, r)}$. Output $\sigma = (\sigma_1, \sigma_2, \sigma_3, s)$.
- **Homomorphic Evaluation:** Given signatures $\Sigma^{(j)} = (\sigma_1^{(j)}, \dots, \sigma_n^{(j)})$ on message vectors $\mathbf{m}^{(j)}$ and scalar coefficients $a_j \in \mathbb{F}_p$, compute a signature on the linear combination $\mathbf{m} = \sum_j a_j \mathbf{m}^{(j)}$ as:

$$\sigma_i = \prod_j \left(\sigma_i^{(j)} \right)^{a_j} \quad \text{for each } i.$$

The derived signature is $\Sigma = (\sigma_1, \dots, \sigma_n)$.

- **Verify:** On the input of the public key pk , a tag τ , a signature $\sigma = (\sigma_1, \sigma_2, \sigma_3, s)$ and a vector of coefficients $(c_1, \dots, c_k)$, they first verify the tag signature (σ_1, σ_2) over the $\mathcal{S}.\text{Verify}(\text{pk}_{\mathcal{S}}, \tau)$. Now check that each coordinate signatures are correct. To do that, they take use of a quick pairing check using bilinear pairings. We refer the details in Section 4 of [Fre12] for more details of the verification and the correctness proof.

3.1.8 Libert-Peter-Joye-Yung scheme with Homomorphic structure preservation [LPJY15]

Libert et al. introduce the first linearly homomorphic structure-preserving signature (SPS) scheme. The key novelty is that all components of signatures, public keys, and verification equations consist solely of group elements, without relying on hash functions, encoding, or random oracles. The scheme supports linear combinations of signed messages and achieves adaptive unforgeability and strong context hiding in the standard model, making it particularly well suited for applications in verifiable computation and privacy-preserving data aggregation.

- **Keygen:** Given a dimension n and bilinear group $\mathbb{G}, \mathbb{G}_T$ of prime order p , choose $g_z, g_r, h_u, h_z \xleftarrow{\$} \mathbb{G}$. Also pick $\chi_i, \gamma_i, \delta_i \xleftarrow{\$} \mathbb{Z}_p$. Then compute $g_i = g_z^{\chi_i} g_r^{\gamma_i}$ and $h_i = h_z^{\chi_i} h_u^{\delta_i}$. The secret key is $\text{sk} = \{\chi_i, \gamma_i, \delta_i\}_{i=1}^n$ while the public key is $\text{pk} = (g_z, g_r, h_z, h_u, \{g_i, h_i\}_{i=1}^n)$
- **Sign:** To sign a vector $(m_1, \dots, m_n) \in \mathbb{G}^n$, associated with some identifier τ (can be left undefined in this case, as it is not involved in the scheme), the signer computes the signature $\sigma = (z, r, u) \in \mathbb{G}^3$ where

$$z = \prod_{i=1}^n m_i^{-\chi_i}, \quad r = \prod_{i=1}^n m_i^{-\gamma_i}, \quad u = \prod_{i=1}^n m_i^{-\delta_i},$$

- **Homomorphic Evaluation:** Given the public key pk and ℓ tuples $(\omega_i, \sigma_i = (z_i, r_i, u_i))$. Compute and return the derived signature

$$\sigma = (z, r, u) = \left(\prod_{i=1}^{\ell} z_i^{\omega_i}, \prod_{i=1}^{\ell} r_i^{\omega_i}, \prod_{i=1}^{\ell} u_i^{\omega_i} \right)$$

- **Verify:** Given a signature $\sigma = (z, r, u)$, a vector $m_1, \dots, m_n$, return ‘ok’ if and only if $(m_1, \dots, m_n) \neq (1_{\mathbb{G}_T}, \dots, 1_{\mathbb{G}_T})$ and (z, r, u) satisfy

$$1_{\mathbb{G}_T} = e(g_z, z) \cdot e(g_r, r) \cdot \prod_{i=1}^n e(g_i, m_i) \quad \text{and} \quad 1_{\mathbb{G}_T} = e(h_z, z) \cdot e(h_u, u) \cdot \prod_{i=1}^n e(h_i, m_i)$$

3.1.9 Fuchsbauer-Hanser-Slamanig scheme with Homomorphic structure preservation [FHS19]

This scheme provides a variant of the scheme proposed by Libert et al. in [LPJY15], where the tag is no longer a singleton value ϵ , but rather a randomised vector which provides anonymity (something that was not possible before). The scheme follows the following structure.

- **Keygen:** Let $\mathbb{G}_1, \mathbb{G}_2$ and $\mathbb{G}_T$ be bilinear groups with order p and let g and $\mathbf{g}$ be the generators of $\mathbb{G}_1$ and $\mathbb{G}_2$ respectively. Let $\mathcal{T}$ be the set of tags. One randomly chooses $\text{sk} = (s_1, \dots, s_n) \xleftarrow{\$} \mathbb{Z}_p$ and the public key is defined as $\text{pk} = (\mathbf{g}^{s_i})_{i=0}^n$ for $\mathbf{g}_i = \mathbf{g}^{s_i}$ and $\mathbf{g}_0 = \mathbf{g}$.
- **Sign:** Unlike the previous scheme, the first step is to generate a tag. This is done by first choosing some random $r \xleftarrow{\$} \mathbb{Z}_p$ and set $\tau = (\tau_1 = g^{1/r}, \tau_2 = \mathbf{g}^{1/r})$ and $\tilde{\tau} = r$. For a vector message $(m_1, \dots, m_n) \in \mathbb{G}_1^n$ together with some secret tag $\tilde{\tau}$, one sets the signature $\sigma = (\prod_i m_i^{s_i})^{\tilde{\tau}}$ and outputs (τ, σ) .

- **Homomorphic Evaluation:** Given the public key $\mathbf{pk}$ and ℓ tuples (ω_i, σ_i) . Compute and return the derived signature $\sigma = \prod_{i=1}^{\ell} \sigma_i^{\omega_i}$.
- **Verify:** Given the public key $\mathbf{pk} = (\mathbf{g}_i)_i$, a vector message $(m_1, \dots, m_n)$ and a signature σ under the tag $\tau = (\tau_1, \tau_2)$, first verify if the tag holds by checking $e(\tau_1, \mathbf{g}_0) = e(g, \tau_2)$ followed by $e(\sigma, \tau_2) = \prod_{i=1}^n e(m_i, \mathbf{g}_i)$.

3.1.10 Guo-Liu-Tian-Zheng scheme with Adaptively secure lattice based signatures [GTLZ24]

This work present a linearly homomorphic signature scheme based on standard lattice assumptions, addressing the lack of lattice-based constructions achieving both adaptive security and strong context hiding. Unlike earlier lattice-based schemes that were selective or depth-limited, this work provides a framework supporting adaptive chosen-message security while retaining linear homomorphism. The scheme operates in the standard model and is suitable for post-quantum settings.

- **Keygen:** Let n be the security parameter and $k = \text{poly}(n)$ be the maximum dataset size (and is even). Let there be a prime $q \geq (nk)^2$ and $h = \lfloor \frac{n}{6 \log q} \rfloor$ and $V = \sqrt{2nk \log q} \cdot \log n$. The signer generates $(\mathbf{A}, T_A) \leftarrow \text{TrapGen}(h, q)$. Sample $(\alpha_1, \dots, \alpha_k) \xleftarrow{\$} \mathbb{Z}_q^h$ that are linearly independent. Output the public key $\mathbf{pk} = (\mathbf{A}, \alpha_1, \dots, \alpha_k)$ and the secret key $\mathbf{sk} = T_A$.
- **Sign:** On the input of the secret key $\mathbf{sk}$, the tag $\tau \in \{0, 1\}^n$ and the messages $m_1, \dots, m_k \in (\mathbb{F}_q^k)^k$. Construct a orthogonal matrix $\mathbf{H}_\tau = \text{diag}(2\tau_1 - 1, \dots, 2\tau_n - 1)$ (where $\tau_1, \dots, \tau_n$ represents all the possible tags). Compute a new basis T_{B^τ} for the lattice $\Lambda_\tau = \Lambda_q^\perp(\mathbf{A}\mathbf{H}_\tau)$. This is followed by decomposition⁴ of the message vector m_i into u_i and v_i . Let $t(u_i) = \sum_{j=1}^k u_{ij} \alpha_j$ and $t(v_i) = \sum_{j=1}^k v_{ij} \alpha_j$ and compute $\sigma(u_i)$ and $\sigma(v_i)$ as

$$\sigma(u_i) \leftarrow \text{SamplePre}(\mathbf{A}\mathbf{H}_\tau, T_{B^\tau}, t(u_i), \delta)$$

$$\sigma(v_i) \leftarrow \text{SamplePre}(\mathbf{A}\mathbf{H}_\tau, T_{B^\tau}, t(v_i), \delta)$$

Finally output the signature $\sigma = \sigma(u_i) + \sigma(v_i)$.

- **Homomorphic Evaluation:** Given a set of signatures $\sigma_1, \dots, \sigma_k$ and linear function $\langle f \rangle = (c_1, \dots, c_k)$, it returns

$$\sigma = \sum_i c_i \sigma_i.$$

- **Verify:** On the input of the public key $\mathbf{pk}$, a binary vector τ , a message m , first compute the matrix $\mathbf{H}_\tau$ as before and then compute $\mathbf{t} = \sum_{i=1}^k c_i t_i$ where $t_i = \sum_{j=1}^k m_{ij} \alpha_j$. Then output ok if all the following conditions hold

$$\|\sigma\| \leq kV\sqrt{2nk} \quad \text{and} \quad \mathbf{A}\mathbf{H}_\tau \cdot \sigma = \mathbf{t}$$

3.1.11 Guo-Liu-Tian-Zheng scheme of Linearly Semi-group Homomorphic Signatures [GTLZ25]

This work introduces a homomorphic signature construction based on semigroup actions, providing an alternative algebraic foundation to pairing-based and lattice-based schemes. The core idea is to model homomorphic evaluation as the public composition of semigroup

⁴This is a special function that decomposes the vector into two parts such that the ℓ_2 norm of each are either odd or even. For more details we refer to [GTLZ24, Theorem 8]

actions, allowing signatures to be derived by algebraic combination rather than through bilinear maps or lattice trapdoors. This abstraction enables linearly homomorphic authentication while maintaining constant-size signatures and avoiding the use of random oracles. The scheme achieves adaptive unforgeability and strong context hiding in the standard model under assumptions related to the hardness of forging semigroup action relations. By decoupling homomorphic authentication from specific number-theoretic or lattice structures, this work demonstrates that HS can be realized in a more general algebraic framework, thereby broadening the design space of homomorphic authentication mechanisms. Both the message spaces and signature spaces are both semigroups without inverses and supporting an operation $||$.

- **Keygen:** Let n be the security parameter and $k = \text{poly}(n)$ be the maximum dataset size (and is even). Let there be a prime $q \geq (nk)^2$ and $h = \lfloor \frac{n}{6 \log q} \rfloor$ and $V = \sqrt{2nk \log q} \cdot \log n$. The signer generates $(\mathbf{A}, T_A) \leftarrow \text{TrapGen}(h, q)$. Select a secure collision resistant hash function $H : \mathbb{Z}_+ \mapsto \{0, 1\}^k$ and additionally sample $(\alpha_1, \dots, \alpha_n) \xleftarrow{\$} \mathbb{Z}_q^h$ that are linearly independent. Output the public key $\text{pk} = (\mathbf{A}, H, \alpha_1, \dots, \alpha_k)$ and the secret key $\text{sk} = T_A$.
- **Sign:** On the input of the secret key sk and the message $(m_1, \dots, m_k) \in \mathbb{Z}_+$. Compute $v_i = \mathbf{h}(m_i) \in \mathbb{Z}_q^h$ and $\beta_i = \sum_{j=1}^k v_{ij} \alpha_j \pmod q$ for $i \in [1, k]$. Finally compute the pre-image $\sigma_i \leftarrow \text{SamplePre}(\mathbf{A}, T_A, \beta_i, \delta)$. Output the signature $\sigma = \sigma_1 \sigma_2 \dots \sigma_k \in \mathbb{Z}^n$.
- **Homomorphic Evaluation:** Due to the properties of the semigroup, given two messages $m_1, m_2 \in \mathbb{Z}_+$ we have

$$\text{Sign}(\text{pk}, m_1 || m_2) = \text{Sign}(\text{pk}, m_1) || \text{Sign}(\text{pk}, m_2)$$

- **Verify:** On the input of the public key pk and the message $(m_1, \dots, m_n)$ and the signature σ , first compute $\beta_i = \sum_k (\mathbf{h}(m_i))_j \alpha_j$. Then output ok if all the following conditions hold

$$||\sigma|| \leq V\sqrt{nk} \quad \text{and} \quad \mathbf{A} \cdot [\sigma_1, \dots, \sigma_k] = [\beta_1, \dots, \beta_k].$$

Remark 3. Early constructions of linear HS (prior to 2010) can be interpreted as Fiat–Shamir transforms of Σ -protocols proving linear relations. In particular, the scheme of Boneh et al. [BFKW09] authenticates vectors by generating non-interactive proofs of membership in a signed linear subspace (using the tags τ), with homomorphic evaluation corresponding to linear combination of witnesses (the signatures σ themselves). These Σ -protocol-based schemes are conceptually elegant but typically rely on the random oracle model and are limited to linear functions, motivating later algebraic and lattice-based approaches.

3.2 Non-linear Homomorphic Signatures

While linear HS allow the verification of computations that can be expressed as linear combinations of signed data, their expressiveness is inherently limited to linear functions. Many practical applications, however, require the verification of more complex, non-linear transformations, such as polynomial evaluation, aggregation functions, or even general arithmetic circuits. To address this limitation, the study of non-linear HS extends the homomorphic property beyond linearity, enabling authenticated computations on signed data under more expressive classes of functions.

Non-linear HS permit the derivation of a valid signature on the output $y = f(m_1, \dots, m_k)$ for arbitrary or restricted non-linear functions f , given only the signatures on the original

Table 1: Comparison of linearly homomorphic signature schemes where we make a combination of n messages. For pairings based schemes, g.e. represents group elements.

Scheme	Hard Prob	Sign. size	Security	Privacy	Model
[BFKW09]	DLP	1 g.e.	Selective	Strong	ROM
[GKKR10]	RSA	1 g.e.	Adaptive	None	ROM
[BF11b]	Lattice	$O(n \log^2 q)$	Adaptive	Weak	ROM
[BF11a]	Ideal Lattice	$O(n \log q)$	Adaptive	None	ROM
[AL11b]	Pairings	3 g.e.	Adaptive	Weak	SM
[CFW11]	RSA	1 g.e. + ℓ integers	Adaptive	None	SM
[ALP12]	Pairings	2 g.e.	Selective	Complete	SM
[CFW12]	Sq-DH	1 g.e.	Adaptive	None	SM
[ALP13]	Pairings	6 g.e.	Selective	Complete	SM
[WHW13]	Lattice	$m \log q + n$	Selective	Weak	ROM
[LPJY15]	Pairings	3 g.e.	Adaptive	Strong	SM
[FHS19]	Pairings	1 g.e.	Adaptive	Strong	SM
[GTLZ24]	Lattice	$O(n)$	Adaptive	Weak	SM
[GTLZ25]	Lattices	$O(n)$	Adaptive	Strong	SM

messages $(m_1, \dots, m_k)$. In this setting, the evaluation algorithm **Eval** takes as input the public key, the function description, and the input signatures, and outputs a derived signature σ that can be publicly verified to confirm that y is indeed the correct result of applying f to legitimately signed inputs. Such signatures combine authenticity and verifiability with function privacy, offering a powerful primitive for verifiable computation and secure outsourcing of computations over authenticated data.

Early works in this direction sought to extend homomorphic signature schemes to support polynomial functions, which already require non-linear algebraic relations among inputs. Notable constructions include those based on lattice assumptions and homomorphic trapdoor functions, such as the schemes of GVW [GVW15], and later variants that aim to achieve context hiding and compactness for more general classes of circuits. These schemes differ substantially from linear ones in both structure and security proofs, requiring mechanisms to maintain correctness and unforgeability under multiplicative relations among messages.

In the following subsections, we review the principal non-linear homomorphic signature constructions, their underlying hardness assumptions, and the function classes they support, ranging from low-degree polynomials to bounded-depth arithmetic circuits.

3.2.1 Boneh-Freeman Scheme for Polynomial functions [BF11a]

This scheme was proposed as generalization of the linearly homomorphic scheme presented in Section 3.1.4. To extend this idea to non-linear computations, the authors reinterpret the underlying lattice as an *ideal lattice*, thereby endowing it with a ring structure that supports both addition and multiplication. Note that the security of the scheme still relies on the SIS problem.

Definition 17 (Ideal Lattice). An *ideal lattice* is a lattice that is also an ideal of a ring, typically the ring of integers of a polynomial ring modulo an integer modulus. Formally, let $R = \mathbb{Z}[x]/(f(x))$ be a polynomial ring modulo a monic polynomial $f(x)$, and let $q \in \mathbb{Z}$ be a modulus. A lattice $\Lambda \subseteq \mathbb{Z}^n$ is called an *ideal lattice* if there exists an ideal $I \subseteq R$ such

that $\Lambda = \Phi(I)$, where Φ is the canonical embedding of R into $\mathbb{R}^n$.

Remark 4. The multiplicative structure of R then induces a multiplication operation on the lattice points, making it possible to define products of lattice elements in a way that is consistent with ring multiplication in R . This additional structure is what enables polynomial (non-linear) homomorphic evaluation.

Definition 18 (Prime Ideals). A *prime ideal* $\mathfrak{p}$ of a ring R is an ideal such that for any $a, b \in R$, if $ab \in \mathfrak{p}$ then either $a \in \mathfrak{p}$ or $b \in \mathfrak{p}$. Additionally, $R/\mathfrak{p}$ is a finite field $\mathbb{F}_{p^e}$ where p is the characteristic of $\mathfrak{p}$ and $e \in \mathbb{Z}$ is the degree of $\mathfrak{p}$.

Remark 5. Prime ideals play an important role in the decomposition of ideal lattices, as they correspond to factors of the modulus in the ring R . When R is a quotient of a polynomial ring, prime ideals define the components of the Chinese Remainder decomposition $R/\mathfrak{p}_1 \times \cdots \times R/\mathfrak{p}_t$, which in turn allows parallel or component-wise evaluation of polynomial functions on signatures, a property exploited in homomorphic constructions.

Theorem 4 ([Ste08] Theorem 8.2). Let $R = \mathbb{Z}[x]/(f(x))$ be a ring and $\mathfrak{p}$ be a prime ideal of R . Then one can write $\mathfrak{p} = p \cdot R + h(x) \cdot R$ for some polynomial $h(x)$, such that $h(x) \bmod p$ is an irreducible factor of $f(x) \bmod p$. One can also represent the prime ideal $\mathfrak{p}$ as $(p, h(x))$ in a two-element representation.

In Boneh and Freeman's scheme, the ideal lattice representation ensures that both message and signature spaces inherit the ring's multiplicative structure. By performing arithmetic on the ring elements corresponding to the lattice points, the homomorphic evaluation of polynomial functions becomes natural and algebraically consistent with the original linear scheme, while maintaining security under standard lattice assumptions.

Let $\ell = \binom{k+d}{d} - 1$. Let $\{Y_j\}_{j=1}^\ell$ be the set of all non-constant monomial $x_1^{e_1} \cdots x_k^{e_k}$ of degree $\sum e_i \leq d$, in lexicographic order. Any polynomial function f that belongs to the admissible set of functions is defined by $f(\mathbf{m}) = \sum_{j=1}^\ell c_j Y_j(\mathbf{m})$ for $c_j \in \mathbb{F}_p$.

- **Keygen:** Consider a monic irreducible polynomial $F \in \mathbb{Z}[x]$ of degree n . Using algorithms from [SV10] generate two distinct principle degree one prime ideals from ring R , $\mathfrak{p} = (p, x - a)$ and $\mathfrak{q} = (q, x - b)$ with generators g_p, g_q , respectively. Let $T = \{g_p g_q, g_p g_q x, \dots, g_p g_q x^{n-1}\}$ be a basis of $\mathfrak{p} \cdot \mathfrak{q}$. Assuming that we have the hash function (as defined in Section 3.1.4), output the public key $\mathbf{pk} = (f, p, q, a, b, \delta, H)$ and secret key $\mathbf{sk} = T$. The evaluation of the hash function is modeled as a random oracle.
- **Sign:** On the input of the secret key, a tag $\tau \in \{0, 1\}^n$, a message $m \in \mathbb{F}_p$ and an index i , compute $\alpha_i = H(\tau || i)$. Compute $h(x) \in R$ such that $h(a) \bmod p = m$ and $h(b) \bmod q = \alpha_i$. Finally output, the signature as an element of the ideal $\mathfrak{p} \cdot \mathfrak{q}$ using

$$\sigma \leftarrow \text{SamplePre}(\mathfrak{pq}, T, h, \delta)$$

- **Homomorphic Evaluation:** On the input of public key $\mathbf{pk}$, tag τ , a function f encoded as $\langle f \rangle = (c_1, \dots, c_\ell)$ and signatures $\sigma_1, \dots, \sigma_k \in \mathbb{Z}^n$

1. Lift $f \in \mathbb{F}_p[x_1, \dots, x_k]$ to $\mathbb{Z}[x_1, \dots, x_k]$ by setting $\hat{f} = \sum_{j=1}^\ell c_j Y_j(x_1, \dots, x_k)$.
2. Output $\hat{f}(\sigma_1, \dots, \sigma_k)$.

- **Verify:** For the input of public key, a tag τ , a message $m \in \mathbb{F}_p$, a signature $\sigma \in R$ and an acceptable function f , accept only if $\|\sigma\| \leq L$, where L is a certain bound and

$$\sigma(a) \bmod p \equiv m \quad \text{and} \quad \sigma(b) \bmod q \equiv \omega_\tau(\langle f \rangle)$$

Note 2. When R is viewed as an additive group, the scheme coincides with the linear homomorphic signature. By reinterpreting R as a ring and endowing the lattice with multiplicative closure, the construction naturally generalizes to support polynomial evaluations.

3.2.2 Hiromasa-Manabe-Okamoto Scheme over Polynomial functions with Shorter Signatures [HMO13]

This signature scheme was proposed with the idea of reducing the signature sizes of the Boneh-Freeman scheme [BF11a]. The aforementioned scheme uses the preimage sampling [GPV08], however, this work uses an improved preimage sampling algorithm [MP12] to achieve a significantly smaller signatures, albeit with some increase in the length of the secret key. The construction of this scheme is the same as [BF11a], however, a new trapdoor generator $\text{TrapGen}'$ (Algorithm 1, [HMO13]) using the basis T is used to generate a matrix $\mathbf{H} \in \mathbb{Z}^{n \times n}$ and a trapdoor $\mathbf{R}^5$ (based on a choice of matrix, $\mathbf{A} = (a, ax, \dots, ax^{n-1}) \in \mathbb{Z}^{n \times n}$) such that the following holds

$$\mathbf{A}(\mathbf{R} || I_n) = \mathbf{H}\mathbf{G}.$$

The construction is as follows

- **Keygen:** The process follows exactly the same step except that one runs the $\text{TrapGen}'(T)$ where T is the basis as defined in 3.2.1 to obtain $\mathbf{A}, \mathbf{H}, \mathbf{R}$. The public key is the same as before but the $\text{sk} = (\mathbf{A}, \mathbf{R}, \mathbf{H})$.
- **Sign:** On the input of the secret key, a tag $\tau \in \{0, 1\}^n$, a message $m \in \mathbb{F}_p$ and an index i , compute $\alpha_i = H(\tau)$. Compute $h(x) \in R$ such that $h(a) \bmod p = m$ and $h(b) \bmod q = \alpha_i$. Identify a ring element h that satisfies $\mathbf{A}h = u \bmod q$ and output the signature

$$\sigma \leftarrow \text{SamplePre}'(A, R, H, u, \delta)^6$$

- **Homomorphic Evaluation:** On the input of public key pk , tag τ , a function f encoded as $\langle f \rangle = (c_1, \dots, c_\ell)$ and signatures $\sigma_1, \dots, \sigma_k \in \mathbb{Z}^n$
 1. Lift $f \in \mathbb{F}_p[x_1, \dots, x_k]$ to $\mathbb{Z}[x_1, \dots, x_k]$ by setting $\hat{f} = \sum_{j=1}^{\ell} c_j Y_j(x_1, \dots, x_k)$.
 2. Output $\hat{f}(\sigma_1, \dots, \sigma_k)$.
- **Verify:** For the input of public key, a tag τ , a message $m \in \mathbb{F}_p$, a signature $\sigma \in R$ and an acceptable function f , accept only if $\|\sigma\| \leq L$, where L is a certain bound and

$$\sigma(a) \bmod p \equiv m \quad \text{and} \quad \sigma(b) \bmod q \equiv \omega_\tau(\langle f \rangle)$$

Using the more efficient algorithm of [MP12], the fresh signature size in this scheme is $\mathcal{O}(n^3)$, while that of [BF11a] is $\mathcal{O}(n^{4.5})$. However this comes with a significant increase in the secret key size from $\mathcal{O}(n)$ to $\mathcal{O}(n^{3.5})$.

Note 3. The Homomorphic evaluation and the Verify algorithm for both [BF11a] and [HMO13] are exactly the same as the goal of the second scheme was to shorten the signature sizes.

3.2.3 Catalano-Fiore-Warinschi Scheme for Polynomial Functions with efficient verifications [CFW14]

Catalano, Fiore, and Warinschi [CFW14] presented an improved construction of HS for polynomial functions, addressing two main limitations of previous schemes such as Boneh and Freeman [BF11a]: (i) the inefficiency of verification, which in earlier constructions required large-lattice computations, and (ii) reliance on heuristic or random-oracle assumptions. Their approach achieves *efficient verification* and standard-model security by adopting algebraic encodings and proof systems that allow verification through a

⁵this matrix is a $\mathbf{G}$ – trapdoor [MP12] which is chosen based on a discrete Gaussian distribution

⁶Note that this uses a modified $\text{SamplePre}'$ algorithm drawing inspiration from [MP12]

small number of group operations, even for derived signatures corresponding to non-linear (polynomial) functions.

The scheme relies on a new problem that was introduced in this paper which has defined informally below.

Definition 19 (k-Augmented-Power Multi-linear Diffie-Hellman [CFW14]). Given level-1 encodings $g_1^a, g_1^b, g_1^{ab}, g_1^x, g_1^{xa}, g_1^{abx}$ with $a, b, x \in \mathbb{Z}_p$, it must be hard to compute a level- k encoding of $a^{k-1}(bx)^k$ (i.e. $g_k^{a^{k-1}(bx)^k}$).

This scheme uses a predefined signature scheme. $\mathcal{S} = (\text{Keygen}, \text{Sign}, \text{Verify})$

- **Keygen:** Generate the keypair $(\text{pk}', \text{sk}') \leftarrow \mathcal{S}.\text{Keygen}$ for the signature scheme $\mathcal{S}$, a set of $2k$ linear groups $\mathbb{G}_1, \dots, \mathbb{G}_{2k}$ of order p . Choose random elements $g_1, h_1 \xleftarrow{\$} \mathbb{G}_1$ and $n + 1$ random values $R_\tau \xleftarrow{\$} \mathbb{G}_1$. Also for an element $g_1(h_1 \text{ resp.}) \in \mathbb{G}_1$, one denotes $g_i(h_i \text{ resp.}) \in \mathbb{G}_i$ the i -fold multiplication of $g_1(h_1 \text{ resp.})$. Finally output the public key as $\text{pk} = (\text{pk}', g_1, h_1, \{R_\tau\})$ and the secret key $\text{sk} = (\text{sk}', K)$ where K is a random seed.
- **Sign:** It takes in the secret key sk , a label τ and message m . The signing is done in two steps. In the pre-signing stage, one generates two integers (a, b) for a data set (using a PRF and seed K), and computes,

$$A_1 = g_1^a, B_1 = g_1^b, C_1 = g_1^{ab}, T_1 = h_1^a \text{ and } U_1 = h_1^{ab}$$

Now compute $\Lambda_1 = (R_\tau h_1^{-m})^b$, and $\Gamma_1 = \Lambda_1^a$ to produce the homomorphic part of the signature $\nu = (m, \Lambda_\tau, \Gamma_\tau)$. To construct the final signature, the parameters of the message and the dataset are signed as well using the signature scheme $\mathcal{S}$ as $\sigma_\tau = \mathcal{S}.\text{Sign}(\text{sk}', A_1, B_1, C_1, T_1, U_1, \nu_\tau)^7$. The final signature is

$$\sigma = (A_1, B_1, C_1, T_1, U_1, \nu_\tau \sigma_\tau, \nu)$$

- **Homomorphic Evaluation:** The evaluation algorithm takes in the public key pk , a function f , a vector of signatures $\sigma^{(1)}, \dots, \sigma^{(n)}$. For evaluation correctness, the signatures must share the same parameters and the σ_τ . The algorithm proceed gate by gate for the function where each gate is fan-in-2. Consider the two inputs $\nu_1 = (m_1, \Lambda_1^{(1)}, \Gamma_1^{(1)})$ and $\nu_2 = (m_2, \Lambda_1^{(2)}, \Gamma_1^{(2)})$. For an addition gate, i.e. $m = m_1 + m_2$ compute $\Lambda_i = \Lambda_i^{(1)} \cdot \Lambda_i^{(2)}$ and $\Gamma_i = \Gamma_i^{(1)} \cdot \Gamma_i^{(2)}$. For a multiplication gate, it computes

$$m = m_1 \cdot m_2$$

$$\Lambda_d = e(\Lambda_i^{(1)}, \Lambda_j^{(2)}) \cdot e(\Lambda_i^{(1)}, U_j^{m_2}) \cdot e(U_i^{m_1}, \Lambda_j^{(2)})$$

$$\Gamma_d = e(\Gamma_i^{(1)}, \Gamma_j^{(2)}) \cdot e(\Gamma_i^{(1)}, U_j^{m_2}) \cdot e(U_i^{m_1}, \Gamma_j^{(2)})$$

- **Verify:** The verification happens in two stages. Using the verification key, verify that the signature of the dataset Δ is valid. If σ_Δ is valid, then use graded maps⁸ to evaluate the circuit f over the R_τ , to get a $R = f(R_{\tau_1}, \dots, R_{\tau_n}) \in \mathbb{G}_d$ and the verify if

$$e(R \cdot h_d^{-m}, g_d^{a^{d-1}b^d}) \equiv e(\Lambda_d, g_d) \text{ and}$$

$$e(\Lambda_d, A_1) \equiv e(\Gamma_d, g_1)$$

⁷The $\nu_\tau = (1, \Lambda_\tau, \Gamma_\tau)$ forms the signature of '1' with respect to the label τ , where $\Lambda_\tau = ((R_\tau h_1^{-1})^b)$ and $\Gamma_\tau = \Gamma_1$. For a single dataset, this can be reused for every message m .

⁸Refer to [CFW14] for more details

The verifier first verifies if the final signature truly represents the homomorphic evaluations of the arithmetic circuits, while the second verifies if the twin signatures still hold the same relations after passing through the homomorphic circuits. The idea to make this verification efficient, which is a major contribution of this work is to pre-compute certain components like $R, h_d, g_d^{a^{d-1}b^d}$ and share them as a part of the public key pk . To convert this selectively secure scheme into an adaptively secure one, the article discusses a method and we refer the readers to [CFW14, Section 3] for more details. Additionally, under a strong interactive variant of the previously discussed problem, the homomorphic signature scheme is by itself adaptive secure.

Note 4. They show how to convert a weakly secure HS that works for arithmetic circuits of degree k into adaptive secure one but only restriction is that message space is expected to be some finite field $\mathbb{Z}_p$ that does not *depend on the secret key*.

Table 2: Comparison of non-linear homomorphic signature schemes

Scheme	Hard Prob	Sign. size	Security	Privacy	Model
[BF11a]	Lattice	$\mathcal{O}(n^{4.5})$	Selective	None	ROM
[HMO13]	Lattice	$\mathcal{O}(n^3)$	Selective	None	ROM
[CFW14]	DLP	$\mathcal{O}(d^3 + d^2 \log n)$	Adaptive	None	SM

3.3 Levelled and Fully Homomorphic Signatures

3.3.1 Gorbunov-Vaikuntanathan-Wichs scheme from standard lattices [GVW15]

This work was one of the first works which proposed a first (levelled) fully homomorphic signature scheme for evaluation of any arbitrary circuit over signed binary message. The scheme relies on the novel notion of Homomorphic trapdoor functions (HDTF) which drew inspirations from techniques of [GSW13, BGG⁺14] while using the hardness of the SIS problem.

The maximal depth d needs to be fixed from before.

Definition 20 (Homomorphic trapdoor function (HDTF)). A homomorphic trapdoor function comprises of five polynomial-time functions ($\text{keygen}, f, \text{Inv}, \text{Eval}^{\text{in}}, \text{Eval}^{\text{out}}$) that permits values $\{u_i, x_i, v_i = f_{pk, x_i}(u_i)\}$ and creates new input u^* and output v^* such that $v^* = f_{pk, \mathcal{G}(x_1, \dots, x_n)}(u^*)$ for some arbitrary circuit $\mathcal{G} : \{0, 1\}^n \mapsto \{0, 1\}$ using

$$u^* = \text{Eval}^{\text{in}}(\mathcal{G}, (u_1, x_1), \dots, (u_n, x_n))$$

$$v^* = \text{Eval}^{\text{out}}(\mathcal{G}, v_1, \dots, v_n)$$

Additionally, the Inv functions is a pre-image sampler such that the distributions $(pk, u, v \leftarrow f_{pk, x}(u))$ and $(pk, u \leftarrow \text{Inv}_{sk, x}(v), v)$ are statistically indistinguishable.

Definition 21 (HTDF based on SIS). Let $\mathcal{V} = \mathbb{Z}_q^{n \times m}$, let $\mathcal{U} = \{\mathbf{U} \in \mathbb{Z}_q^{m \times m} : \|\mathbf{U}\| \leq \beta\}$. Using trapdoor generator TrapGen , keygen produce $(\mathbf{A}, \text{td})$ and output $pk = \mathbf{A} \in \mathbb{Z}_q^{n \times n}$ and $sk = \text{td}$. Define $\mathbf{V} = f_{pk, x}(\mathbf{U}) = \mathbf{A} \cdot \mathbf{U} + x \cdot \mathbf{G}$ and $\mathbf{U} \leftarrow \text{Inv}_{sk, x}(\mathbf{V}) = \text{SamplePre}(\mathbf{A}, \text{td}, \mathbf{V} - x \cdot \mathbf{G})$, where $\mathbf{G} \in \mathbb{Z}_q^{n \times m}$ is a deterministically computable public gadget matrix such that $\mathbf{G}' \cdot \mathbf{R} = \mathbf{V}$. The matrix $\mathbf{R}$ has only 0s and 1s [MP12].

Now, based on these new notion of HDTF, the authors construct a signature scheme. One must note that for schemes based on HDTFs, the depth d must be pre-defined.

- **Keygen:** Generate values $(\mathbf{V}_1, \dots, \mathbf{V}_n) \xleftarrow{\$} \mathcal{V}$ and $(\mathbf{A}, \text{td}) \leftarrow \text{TrapGen}$ and publish $pk = \mathbf{A}$ and store $sk = (\text{td}, \mathbf{V}_1, \dots, \mathbf{V}_n)$.

- **Sign:** Given messages $(x_1, \dots, x_n)$, sample $\mathbf{U}_i \leftarrow \text{Inv}_{sk', x_i}(\mathbf{V}_i)$ and set $\sigma_i = \mathbf{U}_i$. Additionally, compute $\mathbf{V}_G^* = \text{Eval}^{\text{out}}(\mathcal{G}, \mathbf{V}_1, \dots, \mathbf{V}_n)$.
- **Homomorphic Evaluation:** Given signatures and their corresponding messages $((\sigma_1, x_1), \dots, (\sigma_n, x_n))$ and a circuit $\mathcal{G}$ call

$$\sigma^* = \text{Eval}^{\text{in}}(\mathcal{G}, (\sigma_1, x_1), \dots, (\sigma_n, x_n))$$

The algorithms Eval^{in} and Eval^{out} respectively perform the following computations

1. Addition : Let $\mathcal{G}(x_1, x_2) = x_1 + x_2$, then we have

$$\mathbf{U}^* = \mathbf{U}_1 + \mathbf{U}_2 \quad , \quad \mathbf{V}^* = \mathbf{V}_1 + \mathbf{V}_2$$

2. Multiplication : Let $\mathcal{G}(x_1, x_2) = x_1 \cdot x_2$. Let $\mathbf{R} = \mathbf{G}^{-1}(\mathbf{V}_1)$ and $\mathbf{GR} = -\mathbf{V}_1$, then we have

$$\mathbf{U}^* = x_2 \cdot \mathbf{U}_1 + \mathbf{U}_2 \mathbf{G}^{-1}(\mathbf{V}_1) \quad , \quad \mathbf{V}^* = \mathbf{V}_2 \cdot \mathbf{G}^{-1}(\mathbf{V}_1)$$

- **Verify:** Given a signature σ and a message y , a value $\mathbf{V}_G^*$ a circuit $\mathcal{G}$ check if $f_{pk, y}(\sigma) \equiv \mathbf{V}_G^*$ where $f_{pk, y}$ is defined in Definition 21.

Remark 6. This version of the signature has been shown to be selectively secure, however the article also provides a construction to convert this selectively secure homomorphic signature scheme in to a fully secure scheme. The idea is to sign the randomised values $\mathbf{V}_i$ s using the same scheme and include them in the final signature as $\sigma = (\mathbf{U}_i, \mathbf{V}_i, \bar{\sigma}_i)$ where $\bar{\sigma} = \text{Sign}(\mathbf{V}_i)$.

Note 5. There is a difference between the scheme proposed in [BF11a] and this work. For a noise level of β for input, the noise level for a depth d grows up to β^{2^d} for the prior while that this work grows only up to $\beta(m+1)^d$.

3.3.2 Boyen-Fan-Shi scheme for adaptively secure signatures [BFS14]

The article introduces one of the first homomorphic signature constructions that achieves adaptive unforgeability in the standard model. Their work addresses a fundamental limitation of earlier homomorphic signature schemes, which were typically proven secure only in a selective setting, requiring the adversary to commit to the target dataset or computation in advance. This scheme overcomes this restriction by introducing a refined lattice-based design that tightly binds signatures to dataset identifiers while allowing homomorphic evaluation without revealing additional information about the underlying messages. Built from standard lattice assumptions, specifically variants of the SIS problem, the construction supports linearly homomorphic operations with compact signatures and efficient verification. As such, it represents a key step toward practical and robust homomorphic authentication in dynamic and adaptive environments, such as streaming data and incremental computation.

- **Keygen:** Sample $2l + 1$ matrices $\mathbf{A}$ and $\{\mathbf{D}_{ib}\}_{i \in [l], b \in \{0,1\}}$ and t random matrices $\{\mathbf{T}_i\}$ from $\mathbb{Z}^{n \times m}$. Additionally, sample $\mathbf{A}^*, T_A \leftarrow \text{TrapGen}(q, n, m)$. Output the public key $\text{pk} = (\mathbf{A}, \mathbf{A}^*, \mathbf{G}, \{\mathbf{D}_{ib}\}, \{\mathbf{T}_i\})$ and the secret key $\text{sk} = T_{A^*}$.
- **Sign:** Given the secret key sk , a tag τ , a message m and an index i , define a matrix $\mathbf{T} = \sum_{i=0}^{|\tau|} (-1)^{t[i]} \mathbf{T}_i$ where t are the first λ bits of the tag τ . Set the matrix $\mathbf{A}_t = [\mathbf{A}^* | \mathbf{A} + \mathbf{T}]$ and sample the signature as

$$\sigma \leftarrow \text{SamplePre}(\mathbf{A}_t, T_A, \mathbf{D}_{i, b[i]} + m\mathbf{G}, \delta)$$

where b is the last ℓ bits of the τ .

- **Homomorphic Evaluation:** It takes in the the public key $\mathbf{pk}$, a tag τ , a pair of message-signature (m_1, σ_1) and (m_2, σ_2) and the circuit $\mathbf{D}$, the following computations are performed

1. Addition. Let $\mathbf{D} = \mathbf{D}_1 + \mathbf{D}_2$, then we have

$$\mathbf{A}_t \sigma_1 + \mathbf{A}_t \sigma_2 = \mathbf{D}_1 + m_1 \mathbf{G} + \mathbf{D}_2 + m_2 \mathbf{G}$$

$$\mathbf{A}_t \sigma_1 + \mathbf{A}_t \sigma_2 = (\mathbf{D}_1 + \mathbf{D}_2) + (m_1 + m_2) \mathbf{G} = \mathbf{A}_t (\sigma_1 + \sigma_2)$$

2. NAND gate. Assume that we have the NAND gate with inputs (u, v, w) . For each gate assume $\mathbf{D}_i$ be the public matrix associated with the wire. So we have

$$\mathbf{A}_t \sigma_1 = \mathbf{D}_u + x \mathbf{G}, \quad \mathbf{A}_t \sigma_1 = \mathbf{D}_v + y \mathbf{G}$$

where x, y are the values carried on wires u, v . Define the output wire public matrix $\mathbf{D}_w = \mathbf{D}_v \widetilde{\mathbf{D}}_u$, where $\widetilde{\mathbf{D}}_u = \mathbf{G}^{-1}(\mathbf{D}_u)$, then the output signature is given by

$$\sigma = \sigma_2 \widetilde{\mathbf{D}}_u - y \sigma_u$$

- **Verify:** Given the public key $\mathbf{pk}$, a tag τ a message m and a signature σ , and the public matrix associated with the polynomial circuit $\mathbf{D}$, compute the $\mathbf{T} = \sum_{i=0}^{|\mathbf{t}|} (-1)^{t[i]} \mathbf{T}_i$ and $\mathbf{A}_t = [\mathbf{A}^* | \mathbf{A} + \mathbf{T}]$. Next verify of the following condition holds

$$\mathbf{A}_t \sigma = \mathbf{D} + m \mathbf{G} \quad \text{mod } q$$

Although the scheme claims support for the evaluation of arbitrary circuits, this should be understood in the sense that any Boolean circuit can be compiled into a linear computation over a suitably prepared authenticated dataset. The scheme does not provide algebraic homomorphism for multiplication gates, nor does it support polynomial or circuit evaluation in the fully homomorphic sense. Consequently, the construction remains a linearly homomorphic signature scheme, albeit with enhanced expressive power through circuit linearization.

Table 3: Comparison of fully homomorphic signature schemes

Scheme	Hard Prob	Sign. size	Security	Privacy	Model
[GVW15]	Lattice	$\text{poly}(d)$	Selective	Weak	SM
[GVW15]	Lattice	$\text{poly}(d)$	Adaptive	Weak	ROM
[BFS14]	Lattice	$\mathcal{O}(n)$	Adaptive	None	SM

3.4 Transition to Proof based Signatures

3.4.1 Gennaro-Wichs SNARK based construction using FHE [GW13]

The scheme represents the first universal approach to homomorphic authentication. The core idea is to decouple the authentication of the data from the computation on that data. In the symmetric setting, this is achieved by encrypting messages within an FHE scheme and applying a standard MAC to the ciphertexts. To move from this private-key setting to a publicly verifiable Homomorphic Signature, the evaluator performs the computation on encrypted or committed data and then generates a zk-SNARK to prove that the evaluation was performed correctly on authenticated inputs. This established the Commit-and-Prove blueprint that defines the modern state-of-the-art.

- **Keygen:** Consider a homomorphic encryption scheme FHE and generate a keypair $\text{pk}_{\text{FHE}}, \text{sk}_{\text{FHE}}$. Similarly for some standard signature scheme Sig generate a keypair $\text{pk}_{\text{Sig}}, \text{sk}_{\text{Sig}}$.
- **Sign:** Given a message m_i for some tag τ , perform the encryption to produce the ciphertext c_i and compute a signature $\sigma_i = \text{Sign}(\text{sk}_{\text{Sig}}, (c_i, \tau))$ and output (c_i, σ_i) .
- **Homomorphic Evaluation:** Given a circuit and a set of ciphertext-signature pairs $\{(c_i, \sigma_i)\}_{i=1}^k$, the evaluator performs the homomorphic computation to compute the resulting ciphertext. Additionally, the evaluator generates a succinct proof π that attests the following : The ciphertexts $(c_1, \dots, c_k)$ and the valid signatures $(\sigma_1, \dots, \sigma_k)$ such that $\text{Verify}(\text{pk}_{\text{Sig}}, (c_i, \tau), \sigma_i) = \text{ok}$ and homomorphic evaluation of the ciphertext is a correct.
- **Verify:** The verifier verifies that the signature corresponds to the final message as well as checks the succinct proof π , that mathematically guarantees that the ciphertext is a result of the authorized homomorphic computation.

This work was revolutionary for its universality, yet it faced two primary criticisms, the heavy computational overhead of FHE and the reliance on non-falsifiable Knowledge assumptions required for the zk-SNARKs. These limitations catalysed the search for more efficient alternatives, leading eventually to the development of lattice-based HS and the pairing-based Functional Commitment approach popularized by [CFT22], which replaced FHE and SNARKs with more efficient, algebraic Commit-and-Prove modules.

3.4.2 Catalano-Fiore-Tucker scheme from additive Functional Commitments [CFT22]

The core philosophy of the construction is a modular separation of duties, a Functional Commitment handles the complex algebraic logic (polynomials or span programs), while a linearly homomorphic signature handles the multi-input authentication. By signing the commitments rather than the messages, the signer creates a structure where an evaluator can homomorphically combine both the data and the authenticity proofs simultaneously. This specific instantiation utilizes the Randomness-Reuse technique to sign group elements, providing a highly efficient pairing-based system that is secure against adaptive adversaries without the need for heavy generic compilers.

- **Global setup and Keygen:** Let $\mathcal{G}$ be the bilinear groups with order p and let g_1, g_2 be the generators of the groups. Generate the CRS for the additive homomorphic functional commitment. The signer picks some random $\text{sk} = x \in \mathbb{Z}_p$ and sets the public key $\text{pk} = g_1^x$. The FC provides a set of commitment generators $\mathbf{h} = (h_1, \dots, h_n) \in \mathbb{G}_1^n$.
- **Sign:** Given a message m for some dataset τ , the signer picks some random $r_\tau \in \mathbb{Z}_p$ and then computes a base signature (using a some standard signature scheme) on the dataset. This r acts as the ephemeral key for all the messages in the dataset denoted by τ . The signer then computes the functional commitment $C_i = h_1^{m_i} h_2^{r_\tau}$ for the message m_i (this structure may change depending on the FC structure). The intermediate signature σ'_i is computed as follows:

$$\sigma'_i = (H(\tau) \cdot C_i)^{r_\tau x}$$

Output the final signature $\sigma = (\tau, i, g_2^{r_\tau}, C_i, \sigma'_i)$.

- **Homomorphic Evaluation:** Given a function f in the list of acceptable functions, represented with coefficients $(\alpha_1, \dots, \alpha_k)$ and signature $\sigma_1, \dots, \sigma_k$, the evaluator computes the aggregated commitment C^* and signature σ'

$$C^* = \prod_{i=1}^k C_i^{\alpha_i}, \quad \sigma' = \prod_{i=1}^k (\sigma_i')^{\alpha_i}$$

Additionally, the evaluator also runs FC.Eval to produce a succinct opening proof π that demonstrates $\text{Open}(C^*) \rightarrow y = f(m_1, \dots, m_k)$.

- **Verify:** The verifier performs the following check

$$e(\sigma', g_2) \stackrel{?}{=} e\left(\prod_{i=1}^k H(\tau_i)^{\alpha_i}, e(g_2^r, \text{pk})\right)$$

Additionally, the verifier also confirms that C^* opens to y under the function f by validating the proof π against the CRS.

The paper specifies that by restricting the coefficients such that $\alpha_i \in \{0, 1\}$ and $\alpha_i \neq 0$ for all queried inputs, the scheme inherently avoids Type-III forgeries. This forgery occurs when an adversary creates a valid signature for a new linear combination that effectively cancels out a signer's original contribution. By disallowing zero-coefficients on queried data, the construction remains strongly adaptive without requiring the more computationally expensive [CFN18] security transformation.

3.4.3 Gay-Ursu unlevelled scheme from fully HS [GU24]

This paper addresses a long standing issues in authenticated computation : constructing an unlevelled fully homomorphic signature. As seen in Section 3.3, the constructions were strictly levelled, i.e. the maximum depth of the circuit must be known at setup, or the previously described schemes which relied on non-falsifiable assumption like knowledge of exponents. This work relies on indistinguishability obfuscation (iO) [BGI⁺12, GGH⁺16]. Zero-knowledge hides the values while the iO hides computation structure.

Definition 22 (Indistinguishability Obfuscation). An obfuscator iO for a circuit ensures that for a circuit class $\mathcal{C}$, any two circuits $C_0, C_1 \in \mathcal{C}$ of the same size that compute the same function, the distribution of $\text{iO}(C_0)$ and $\text{iO}(C_1)$ are computationally indistinguishable.

The scheme leverages iO, FHE and zero-knowledge proofs as building blocks where the iO is used to hide evaluation key structure and to proof generation logic to resist adaptive attacks.

- **Keygen:** Using the FHE scheme generate a keypair $(\text{pk}_{fhe}, \text{sk}_{fhe})$ and generate two puncturable random functions keys (K_1, K_2) that are used to derive actual signing tokens. Next two obfuscated programs are created, $\text{Obf}_{\text{GenCRS}}$ and Obf_{Eval} . The first allows for generation of a common random string tailored to a circuit ⁹, while the second allows the signatures to be updated through NAND gates. The public key and the secret key are defined as

$$\text{pk} = (\text{pk}_{fhe}, \text{Obf}_{\text{GenCRS}}, \text{Obf}_{\text{Eval}}), \quad \text{sk} = (K_1, K_2, \text{sk}_{fhe}).$$

The keys K_1, K_2 allow for performing consistency checks inside the obfuscated programs. This designs enables arbitrary depth circuit to be evaluated by decomposing into NAND gates. while relying on iO to prevent leakage of signing or evaluation structure.

⁹the GenCRS takes in the key K_1 and generates a simulation crs and trapdoor

- **Sign:** Given a message m and the secret key sk , the signer generates a NIZK proof π of the statement that ciphertext c is a correct FHE encryption of m . The signature is

$$\sigma = (c, \pi, \ell = 0),$$

where ℓ is the current depth of the circuit (for a fresh signature this is 0.)

- **Homomorphic Evaluation:** Given the public key pk , a function f and set of message, signature pairs (m_i, σ_i) , the evaluation proceeds with each NAND gate of the function using Obf_{Eval} . For the two inputs (m_0, σ_0) and (m_1, σ_1) , first verify if the NIZK proofs hold true, i.e., the ciphertexts are correct encryptions. Next evaluate the NAND gate over the ciphertexts (c_0, c_1) and the messages (m_0, m_1) to get c' and m' respectively and compute a new proof π' certifying that c' was is true encryption of m' . The new signature is $\sigma = (c', \pi', \ell + 1)$, where ℓ is the current level of the input signatures.
- **Verify:** Given a message y , a signature $\sigma = (c, \pi, \ell)$, the public key pk and the function f , the verifier computes the CRS of the level ℓ , and also computes the ciphertext $c_f = \text{FHE.Eval}(\text{pk}, f, ct'_1, \dots, ct'_n)$ and then it returns the result of the NIZK verification test of the proof π but over the statement that c_f is the encryption of the message y .

Note 6. The evaluation function only works if the signatures input to the NAND gate are the same level in the circuit. This might not be the case depending upon the circuit. However, we note that this is an implementation issue and we do not further investigate it.

4 Multi-key Homomorphic Signatures (MKHS)

The transition from single-key to multi-key HS represents one of the most significant shifts in the field's history. In the single-key setting, the evaluator essentially performs intra-dataset computations where all data points share a common trust anchor—the signer's public key. The algebraic structure of the signature (whether based on pairings or lattices) is designed to mirror the computation, allowing the signature to evolve alongside the data. However, real-world applications—such as collaborative research or cross-organizational data analysis—require the ability to compute over data originating from multiple independent sources, each using its own unique secret key. This multiple dataset requirement introduces three primary technical hurdles:

- **Key Incompatibility:** Signatures generated under different public keys are naturally incompatible to one another. Standard homomorphic techniques fail because the algebraic relations are tied to specific, distinct parameters. Overcoming this requires a mechanism for Key Aggregation—the algebraic process of merging independent identities into a single representative public key.
- **The Succinctness Trap:** While it is trivial to aggregate signatures by simply concatenating them, this results in a final signature that grows linearly with the number of participants (k). Maintaining a constant-sized signature while merging disparate keys is the central challenge of the multi-key setting.
- **Insider Corruption:** In a multi-key environment, the security model must account for the possibility that some signers may be malicious. The scheme must ensure that a dishonest participant cannot alter the computation to frame an honest user or forge a result.

To solve these issues, the field evolved through two distinct algebraic paths. Early foundational work by Derler and Slamanig [DS16] demonstrated that in pairing-based settings, Key-Homomorphic Signatures could allow for constant-size key aggregation. Conversely, in the lattice-based setting, frameworks like [FMNP16] utilized matrix concatenation to handle multiple keys, though often at the cost of public-key or signature bloat. Eventually, the literature moved away from pure algebraic homomorphisms and toward the Commit-and-Prove paradigm. This shift allowed researchers to use sophisticated tools like Batch Arguments and Functional Commitments to compress multiple independent authentication statements into a single, succinct proof.

4.1 Foundational Multi-key Frameworks

4.1.1 Fiore-Mitrokotsa-Nizzardo-Pagnin scheme [FMNP16]

Fiore et al. introduce the notion of multi-key homomorphic authenticators, extending homomorphic authentication to scenarios in which data is contributed by multiple independent users, each holding a distinct secret key. From the homomorphic signature viewpoint, their construction enables authenticated computation over inputs signed under different public keys, while preserving public verifiability of the computed result. The scheme is designed for arithmetic computations over authenticated data and supports homomorphic evaluation without requiring any coordination or shared secrets among signers. This work formalizes security notions tailored to the multi-key setting and demonstrates that HS can be meaningfully generalized beyond the single-key model. However, as a matter of fact, they extend the construction of the GVW scheme (presented in Section 3.3.1) into a multi key setting.

- **Keygen:** Generate k random matrices $\mathbf{V}_1, \dots, \mathbf{V}_k \xleftarrow{\$} \mathcal{V}$ and for every party with identifier id do

$$(\mathbf{A}_{\text{id}}, \text{td}_{\text{id}}) \leftarrow \text{TrapGen}(q, m, n).$$

Then output $\text{sk}_{\text{id}} = (\text{td}_{\text{id}}, \mathbf{A}_{\text{id}}, \mathbf{V}_1, \dots, \mathbf{V}_k)$, $\text{ek}_{\text{id}} = \mathbf{A}_{\text{id}}$ and $\text{pk}_{\text{id}} = (\mathbf{A}_{\text{id}}, \mathbf{V}_1, \dots, \mathbf{V}_k)$. The work assumes that the function takes in k inputs, hence they generate only k matrices $\mathbf{V}_i$. However, we believe that this can be made more efficient by using a crs to generate as many as required.

- **Sign:** Given a message m , a secret key sk_{id} , a label $\ell = (\text{id}, \tau)$ for the message (this consists of the id of the party who signs and τ is the dataset identifier), and computes $\mathbf{U}_{\text{id}} \leftarrow \text{SamplePre}(\mathbf{A}_{\text{id}}, \text{td}_{\text{id}}, \mathbf{V}_{\ell} - m\mathbf{G})$. Finally output the signature

$$\sigma = (m, z, \mathbf{I}, \mathbf{U}_{\text{id}}, \mathbf{Z}_{\text{id}})$$

where for a fresh signature $z = m$ and $\mathbf{Z}_{\text{id}} = \mathbf{U}_{\text{id}}$ and $z, \mathbf{Z}_{\text{id}}$ changes during homomorphic operations. The $\mathbf{I}$ is set of id that are involved in the creation of the signature. For a fresh signature, $\mathbf{I} = \{\text{id}\}$.

- **Homomorphic Evaluation:** It takes a pair of signatures σ_1, σ_2 and its corresponding evaluation keys ek_1, ek_2 . Note that both addition and multiplication consists of a fan-in-2 gates. To perform addition, compute $\mathbf{U} = \mathbf{U}_1 + \mathbf{U}_2$ and $\mathbf{Z} = \mathbf{Z}_1 + \mathbf{Z}_2$. For multiplication, define $\mathbf{V}_1 = \sum_{\text{id} \in \mathbf{I}_1} \mathbf{A}_{\text{id}} \mathbf{U}_{\text{id}} + m_1 \mathbf{G}$ and set $\mathbf{U} = m_2 \mathbf{U}_1 + \mathbf{U}_2 \cdot \mathbf{G}^{-1} \mathbf{V}_1$ and $\mathbf{Z} = \mathbf{Z}_1 + \mathbf{Z}_2$
- **Verify:** On the input of the public key pk , a message m and its signature $\sigma = (m, z, \mathbf{I}, \mathbf{U}, \mathbf{Z})$, it performs the following checks

1. check if $\mathbf{I}$ belongs to the set of admissible identifiers of parties.

2. (*Offline Phase*) Compute two values $\mathbf{V}^*$ and $\mathbf{V}^+$ proceeding with each gate. For every addition gate perform $\mathbf{V}^* = \mathbf{V}_L^* + \mathbf{V}_R^*$ and $\mathbf{V}^+ = \mathbf{V}_L^+ + \mathbf{V}_R^+$, while for every multiplication gate perform $\mathbf{V}^* = \mathbf{V}_R^* \mathbf{G}^{-1} \mathbf{V}_L^*$ and $\mathbf{V}^+ = \mathbf{V}_L^+ + \mathbf{V}_R^+$ respectively.
3. (*Online Phase*) The verification algorithm parses $\mathbf{U} = \{\mathbf{U}_{id}\}_{id \in \mathbf{I}}$ and $\mathbf{Z} = \{\mathbf{Z}_{id}\}_{id \in \mathbf{I}}$ and checks

$$\sum_{id \in \mathbf{I}} \mathbf{A}_{id} \mathbf{U}_{id} + mG = \mathbf{V}^*,$$

$$\sum_{id \in \mathbf{I}} \mathbf{A}_{id} \mathbf{Z}_{id} + zG = \mathbf{V}^+$$

This scheme is secure against Forgery III and therefore provides Strong Unforgeability. The paper also provides a version of the scheme that extends the idea of a single data multi key Homomorphic signature to a multiple dataset multi-key Homomorphic Signature. We refer the readers to [FMNP16] for further details. However, this scheme has been shown to be forgeable under insider corruption [XN25].

4.1.2 The Matrioska Compiler [FP18]

A significant milestone in the shift toward multi-user homomorphic authentication was the introduction of the Matrioska compiler by Pagnin and Fiore (2017). Rather than proposing a novel signature primitive, Matrioska provides a generic builder that transforms any single-key homomorphic signature into a multi-key variant. The compiler operates on a Commit-and-Prove architecture: it aggregates individual signatures from k different users and creates a NIZK proof. This proof certifies that each partial signature is valid under its respective public key and that the final result is the correct output of the circuit applied to those authenticated inputs.

While highly flexible, Matrioska's reliance on standard NIZKs results in weak succinctness, where the signature size grows linearly ($O(k)$) with the number of users. This limitation directly motivated the recent 2024 breakthroughs that have been discussed in the following sections, which replaced the simple NIZK with more sophisticated Batch Arguments (BARGs) to achieve full succinctness ($\mathcal{O}(1)$ signature size) while maintaining Matrioska's modular architecture.

4.1.3 Lai-Tai-Wong-Chow scheme secure against insider corruption [LTWC18]

Lai et al. introduce the first MKHS scheme secure against insider corruption. This is a substantially stronger threat model than prior multi-key constructions. In this setting, multiple data owners independently sign their inputs under their own public keys, and a third party can homomorphically evaluate a function over all signed data. Crucially, some signing keys may later be corrupted, nevertheless, the adversary should still be unable to forge signatures on values that are not legitimate outcomes of admissible computations. The key conceptual challenge addressed in this work is that, in a multi-key environment, corrupted signers can collude with the evaluator, revealing secret keys and internal randomness. Classical homomorphic signature security notions break down in this model, since the adversary may generate partial signatures or mix honest and corrupted inputs in subtle ways. To overcome this, the construction abandons purely algebraic homomorphism and instead relies on a proof-based approach. Signatures are accompanied by zero-knowledge proofs asserting correct computation. Homomorphic evaluation corresponds to composing proofs, rather than algebraically combining signatures. The scheme is generic, i.e. it can be instantiated from any EUF-CMA secure standard digital signature scheme, and does not require the signature scheme itself to be homomorphic.

Here we describe the generic construction. Consider the digital signature scheme $\mathcal{S}$, let $\mathcal{F}$ be the family of admissible functions.

- **Keygen:** Let each signer generate their own keypair $(pk, sk) \leftarrow \mathcal{S}.\text{Keygen}$
- **Sign:** To sign a message m_i under a dataset τ , the signer performs $\sigma \leftarrow \mathcal{S}.\text{Sign}(sk, (\tau, m_i))$ and returns (m_i, σ) .
- **Homomorphic Evaluation:** Given a function $f \in \mathcal{F}$, a collection of signatures $\{(m_i, \sigma_i)\}_{i=1}^k$, their set of corresponding public keys $\{(pk_i)\}_{i=1}^k$, the evaluator computes $y = f(m_1, \dots, m_k)$. Then it produces a zero-knowledge proof π for the following statements:
 1. $\mathcal{S}.\text{Verify}(pk_i, (\tau, m_i), \sigma_i) = \text{ok}$ for all i in the collection.
 2. $y = f(m_1, \dots, m_k)$.

The homomorphically evaluated signature is $\sigma = (y, f, \tau, \pi)$.

- **Verify:** To verify a homomorphic signature σ , the verifier checks the opening of the proof π , if it holds, then this guarantees (i) all inputs were correctly signed, (ii) the claimed output y is the result of the function f applied in the input messages $(m_1, \dots, m_k)$, (iii) even if some signers are corrupted, the proof enforces correctness.

The adversary may adaptively corrupt the signing keys and may participate in the evaluation, however, it still cannot produce a valid signature on (y, f) unless f is evaluated over honestly signed inputs. Hence corrupted keys do not help since an adversary has to prove correctness. Additionally, the zero-knowledge prevents leakage of inputs.

This work marks an important transition from algebraic HS to proof-carrying HS, and strongly influenced later constructions based on SNARKs and BARGs.

4.1.4 Aranha-Pagnin's Simple and Linear Multi-key scheme [AP19]

Aranha and Pagnin propose a simple and efficient construction of multi-key linearly HS in bilinear groups. Their scheme extends classical pairing-based linear HS to the multi-key setting by combining algebraic homomorphism with lightweight proof techniques. Unlike later proof-heavy constructions based on general zero-knowledge arguments, the scheme relies on structure-preserving signatures and pairing-based verification equations, augmented with compact proofs of consistency. The result is a conceptually simple construction that supports linear functions over vectors signed under different public keys, while maintaining public verifiability and efficiency. The scheme is designed for linear homomorphic evaluation only, and does not aim to support arbitrary circuits or strong context hiding.

- **Keygen:** Given bilinear groups $\mathbb{G}_1, \mathbb{G}_2$ with generators g_1, g_2 respectively, each signer independently generates a keypair $sk_i \xleftarrow{\$} \mathbb{Z}_q^*$ and $pk_i = g_2^{sk_i}$.
- **Sign:** Given a message m , a tag τ , a secret key sk_i , a hash function H , compute the signature as follows

$$\sigma = (H(\tau || i) \cdot g_1^m)^{sk_i}.$$

Output the signature as (σ, m) .

- **Homomorphic Evaluation:** Given a function f which is represented as a vector of coefficients $(\alpha_1, \dots, \alpha_n)$, and a set of signatures given as σ_i , the evaluator linearly combines the signatures as $\sigma = \prod_i \sigma_i^{\alpha_i}$ which corresponds to the signature of the message $\sum_i \alpha_i \cdot m_i$. Output the final signature as $(\sigma, m_1, \dots, m_n)$.

- **Verify:** Given the public-keys of the participants pk_i , the final message m , a signature σ , the verifier computes

$$m' = \sum_i \alpha_i m_i, \quad c = \prod_{j=1}^n e(g_1^{m_j} \cdot \prod_j H(\tau || i))^{\alpha_i}, \text{pk}_i)$$

and verify if $m = m'$ and $e(\sigma, g_2) = c$.

4.2 Modern Succinct MKHS

4.2.1 Anthoine-Balbas-Fiore scheme of fully succinct multi-key HS [ABF24]

This paper solves a major open problem by constructing the first Multi-Key Homomorphic Signature (MKHS) scheme that is both fully succinct and based on standard, falsifiable assumptions. While previous MKHS schemes either had signature sizes that scaled with the number of users or relied on non-standard assumptions (like SNARKs), this work achieves signatures of constant size, independent of the number of inputs and circuit depth—using only the hardness of the SIS and LWE problems. Their technical breakthrough lies in a novel combination of Batch Arguments (BARGs) and Functional Commitments (FCs), allowing a single signature to compress the verification of multiple independent signatures from different keys into one compact proof.

More precisely, the MKHS comprises of a BARG, a EUF-CMA secure signature scheme $\mathcal{S}$, a somewhere extractable commitment EC and a functional commitment FC .

- **Keygen:** Each user independently generates a keypair $(\text{pk}, \text{sk}) \leftarrow \mathcal{S}.\text{Keygen}$. Also for the BARG system, there is also a common reference string crs .
- **Sign:** Given a message m_i from some dataset τ the signer i , uses the signature scheme $\mathcal{S}$ to produce the signature $\sigma_i = \mathcal{S}.\text{Sign}(\text{sk}, (\tau, m_i))$.
- **Homomorphic Evaluation:** Given the input of k messages $(m_1, \dots, m_k)$, their corresponding signature $(\sigma_1, \dots, \sigma_k)$ and a function f , the evaluator first re-commits the messages to produce a constant size string $c = \text{FC.Commit}(m_1, \dots, m_k)$. The evaluator further computes the function f and generates an opening proof π showing that y is the correct result of f applied to the inputs, i.e.

$$\pi_1 \leftarrow \text{FC.Open}(f, c, (m_1, \dots, m_k))$$

However, this still does not prove that for every message m_i in the commitment, there exists a valid signature σ_i under the public key pk_i . Instead of sending k signatures, the evaluator generates one single batched argument proof $\pi_2 = \text{BARG.Prove}(\forall i \in [k] : \exists \sigma_i \text{ s.t. } \text{Verify}(\text{pk}_i, m_i, \sigma_i) = \text{ok})$. Output (y, c, π_1, π_2) .

- **Verify:** The verifier first verifies that the $\text{FC.Verify}(c, f, y, \pi_1) = \text{ok}$, i.e. ensures the homomorphic computation correctness. Following which, the batch arguments are checked to ensure that the messages in the commitment are signed by the k users. I.e.

$$\text{BARG.Verify}(\text{pk}_1, \dots, \text{pk}_k, c, \pi_2) = \text{ok}$$

The use of batched arguments allows the signature not to grow with the number of users.

4.2.2 Afshar-Cheng-Goyal scheme with Context hiding [ACG24]

Afshar et al. propose a fundamentally different approach to HS that departs from the classical algebraic paradigm. Instead of embedding homomorphic structure directly into signatures, their construction treats homomorphic evaluation as a certified computation. The scheme relies on standard digital signatures for authentication, functional commitments to encode data and computation, and succinct batched arguments to certify the correctness of evaluated functions. As a result, homomorphic properties arise not from algebraic manipulation of signatures, but from verifiable proofs that committed values are correctly computed. This design enables strong context hiding, multi-key and multi-hop evaluation, and security in the standard model, while remaining compatible with any EUF-CMA-secure signature scheme.

The scheme comprises of a standard digital signature scheme $\mathcal{S}$, a commitment scheme Com and a BARG as a succinct zero knowledge argument system for circuit evaluation. The scheme at a high level is as follows:

- **Keygen:** Each signer i obtains its own set of $\text{pk}_i, \text{sk}_i \leftarrow \mathcal{S}.\text{Keygen}()$ and publishes the pk_i . The system uses a hash function and additionally, a crs is generated that has the parameters of the BARG and the commitment scheme.
- **Sign:** Given a dataset τ , a message m and a public key pk_i compute a context hash $h_i = H(\tau, \text{pk}_i)$ and generates a signature $\sigma_i = \mathcal{S}.\text{Sign}(\text{sk}_i, (m_i, h_i))$ and output the fresh signature (m_i, σ_i, h_i) .
- **Homomorphic Evaluation:** The evaluator takes a circuit f and inputs signatures $(\sigma_1, \dots, \sigma_k)$. However instead of combining the signatures, it creates a certificate of execution. Using a global hash h the evaluator commits to all inputs and the internal states of the computation. The evaluator also generates two succinct batched arguments
 1. π_{auth} , which proves the inputs inside the hash h have valid signatures,
 2. π_{logic} , which proves that the output y was correctly derived from the application of the function f to the inputs.

A RAM delegation proof [KP16] is added to prove that the set of public keys used in the computation is the authorized one.

- **Verify:** The verifier checks the integrity of the RAM certificate rather than individually verifying the signatures. The proof verification, makes the system context hiding as the original messages are not made public.

Remark 7. In both [ABF24] and [ACG24], homomorphic evaluation is realized as a modular certification process rather than a direct algebraic operation on signature vectors. While users provide standard digital signatures to authenticate data, the evaluator initiates the process by generating a functional commitment to the set of input messages. This commitment acts as a succinct 'anchor' that binds the signed data to the computation. The evaluator then utilizes batched arguments to aggregate multiple correctness statements—including input signature validity, gate-by-gate circuit logic, and multi-key consistency—into a single succinct proof linked to that commitment.

4.2.3 Xu-Nishide scheme with forward unforgeability against signing key leakage [XN25]

The scheme of Xu and Nishide extends lattice-based multi-key HS to address a significant security limitation of prior construction of [FMNP16]. It enabled an untrusted evaluator to compute over data signed under different keys, producing a short certificate of correct

evaluation. However, if even one signing key is leaked, a malicious server can forge a signature on a fabricated result that involves inputs from uncorrupted signers. To fix this, Xu and Nishide define and achieve forward unforgeability under signing key leakage, i.e. even if an adversary obtains all signing keys, it cannot forge signatures on incorrect computations. In particular, unlike [FMNP16], the $\mathbf{V}_\ell$ are not generated randomly rather they are dependent on the messages. The construction remains based on the Short Integer Solution (SIS) problem, preserving post-quantum hardness. We provide a detailed construction of the scheme below.

- **Setup:** Sample a uniformly random matrix $\mathbf{M} \in \mathbb{Z}_q^{n \times m}$.
- **Keygen:** Every party samples a matrix $\mathbf{A}_{\text{id}}$ and its corresponding secret trapdoor td_{id} using

$$(\mathbf{A}_{\text{id}}, \text{td}_{\text{id}}) \leftarrow \text{TrapGen}(n, m, q)$$

- **Sign:** For a party id , and a message m with label $\ell = (\text{id}, \tau)$, firstly sample a *short*¹⁰ matrix $\mathbf{W}_\ell \in \mathbb{Z}_q^{m \times m}$, and compute $\mathbf{V}_\ell = \mathbf{M}\mathbf{W}_\ell + m\mathbf{G}$. This is the evaluation key for the user. Following this compute

$$\mathbf{U}_\ell = \text{SamplePre}(\mathbf{A}_{\text{id}}, \text{td}_{\text{id}}, \mathbf{V}_\ell - m\mathbf{G})$$

Output the signature $\sigma_\ell = (\mathbf{I}, \mathbf{U}_\ell, \mathbf{W}_\ell)$ where $\mathbf{I} = \{\text{id}\}$ as the collection of all the users involved¹¹.

- **Homomorphic Evaluation:** The work assumes that every addition and multiplication is a fan-in-2 gates like [FMNP16]. Consider we have a couple of message-signature pairs (m_1, σ_1) and (m_2, σ_2) and their corresponding evaluation keys $\mathbf{V}_{\ell_1}, \mathbf{V}_{\ell_2}$. To perform addition, i.e. $y = m_1 + m_2$, one does $\mathbf{U} = \hat{\mathbf{U}}_{L,1} + \hat{\mathbf{U}}_{R,2}$ ¹² and $\mathbf{W} = \mathbf{W}_{\ell_1} + \mathbf{W}_{\ell_2}$, $\mathbf{V} = \mathbf{V}_{\ell_1} + \mathbf{V}_{\ell_2}$. Now to compute multiplication, i.e. $y = m_1 \cdot m_2$, one computes $\mathbf{V} = \mathbf{V}_{\ell_2} \mathbf{G}^{-1} \mathbf{V}_{\ell_1}$. One computes $\mathbf{U} = (m_2) \hat{\mathbf{U}}_{L,\text{id}} + \hat{\mathbf{U}}_{R,\text{id}} \mathbf{G}^{-1} \mathbf{V}_L$ and $\mathbf{W} = m_2 \mathbf{W}_L + \mathbf{W}_R \cdot \mathbf{G}^{-1} \mathbf{V}_L$.
- **Verify:** The process proceeds in exactly the same way as [FMNP16] with an additional step of verifying

$$\mathbf{M}\mathbf{W} + y\mathbf{G} = \mathbf{V} \pmod{q}$$

which allows to achieve forward unforgeability even under insider corruption.

Note 7. The scheme proposed here achieves better security than [FMNP16], however for the multi-dataset scenario it lacks efficient verification. The paper proposes a multiple dataset homomorphic signature that allows for reuse of evaluation keys across datasets taking use of a proxy server that interacts with the signers to generate signatures without learning the signed messages.

Table 4: Comparison of the Multi-Key HS

Scheme	Hard Prob	Sign. size	Security	Privacy	Model
[FMNP16]	Lattice	$\mathcal{O}(n \cdot \text{poly}(\lambda))$	Adaptive	None	SM
[LTC18]	depends on $\mathcal{S}$	depends	Adaptive	Weak	ROM
[AP19]	co-CDH	constant	Adaptive	None	ROM
[ABF24]	depends	constant	Adaptive	Strong	SM
[ACG24]	depends	$\mathcal{O}(\text{poly}(\lambda))$	Adaptive	Strong	SM
[XN25]	Lattice	$\mathcal{O}(n \cdot \text{poly}(\lambda))$	Adaptive + Forward	Weak	SM

¹⁰Such that the infinite norm of the matrix is less than some predefined value β

¹¹For a fresh signature this only contains the id of the signer.

¹²The matrix $\mathbf{U}_\ell$ is expanded to $\hat{\mathbf{U}}_{L,\text{id}}$ where the entries are 0 for $\text{id} \notin L$ and $\mathbf{U}_{\text{id}}$ otherwise. It is similarly done for $\hat{\mathbf{U}}_{R,\text{id}}$.

5 The Future

5.1 Emerging trends

5.1.1 Stronger Security models beyond EUF-CMA

The research landscape has moved decisively beyond the constraints of Existential Unforgeability under Chosen Message Attacks (EUF-CMA). Because HS are inherently malleable, contemporary literature focuses on function-aware unforgeability, which permits authorized homomorphic transformations while strictly prohibiting signatures on values outside the scope of admissible computation. The frontier of this field is now defined by a transition from selective and semi-adaptive security toward fully adaptive models, ensuring unforgeability in dynamic environments where adversaries choose datasets and functions incrementally. Furthermore, the emergence of insider corruption and leakage resilient models addresses realistic threats where an adversary may learn partial secret key material or corrupt a subset of signers. These trends, including forward unforgeability, ensure that even a future key compromise cannot invalidate or forge historical authenticated data, marking a shift toward zero-trust security architectures in distributed computing.

5.1.2 Multi-key Homomorphic Signatures

Early constructions of HS assume a single trusted signer who authenticates all input data. While this setting is sufficient for outsourced computation over data from a single source, it is inadequate for many collaborative and distributed scenarios. In practice, data used in authenticated computation often originates from multiple independent parties, each holding its own signing key. This has motivated the study of MKHS, which enable the public evaluation of functions over data signed under different keys, while allowing anyone to verify the correctness of the result.

One of the primary difficulties in multi-key settings lies in defining appropriate unforgeability guarantees. In particular, the adversary may corrupt one or more of the signers and obtain their signing keys, while other signers remain honest. In such a case, a secure MKHS scheme must prevent an adversary from producing a valid signature on an incorrect computation. This insider corruption model has no direct analogue in classical signature security and requires new, carefully tailored definitions.

From a constructional perspective, existing MKHS schemes can be broadly divided into two classes. The first class consists of algebraic constructions, most notably lattice-based schemes, while the second class comprises proof-based constructions, which combine standard digital signatures with cryptographic proof systems, such as zero-knowledge arguments or functional commitments. Multi-key support is achieved by aggregating signatures and proving, in zero knowledge, that each input was signed by the corresponding signer. This approach offers greater flexibility and can support richer function classes and stronger privacy properties, such as context hiding, at the cost of relying on more complex primitives.

Another important consideration in MKHS is verification efficiency. Unlike single-key schemes, verification in the multi-key setting may require checking consistency across multiple public keys. Some constructions aim to keep verification time independent of the number of signers, while others trade efficiency for stronger security or privacy guarantees. Balancing these trade-offs remains an active area of research.

Overall, multi-key HS represent a significant extension of the HS paradigm. They capture realistic data-sharing scenarios and expose new dimensions of the design space, including insider security, key exposure, and verification scalability. As a result, MKHS schemes play a central role in recent work on authenticated computation and continue to motivate the development of stronger security notions and more expressive constructions.

5.1.3 Proof-based and Commitment based constructions

The field is witnessing a significant shift toward modular architectures that decouple data authentication from homomorphic computation. While early schemes relied on the rigid algebraic structure of the signature itself, modern trends favor a two-step paradigm: signers authenticate cryptographic commitments to their data, and evaluators use non-interactive zero-knowledge (NIZK) proofs to certify the correctness of the computation. This modularity allows for the integration of standard digital signatures with general-purpose proof systems, enabling the authentication of far more complex function classes, such as general arithmetic or Boolean circuits, which were previously inaccessible to purely algebraic constructions.

A primary driver of this trend is the demand for Context Hiding as a first-class security goal. Unlike many algebraic schemes that may inadvertently leak linear relationships between inputs, commitment-based architectures leverage the hiding properties of NIZKs to ensure that an evaluated signature reveals absolutely nothing beyond the final output. This convergence of HS with verifiable computation is particularly interesting for multi-key settings; it sidesteps the algebraic bottleneck of aggregating signature objects by aggregating proofs of valid evaluation. Although these systems currently face higher computational overhead and larger signature sizes, the ongoing research into SNARKs and functional commitments is rapidly narrowing the efficiency gap, making privacy-preserving, multi-tenant authenticated computation a practical reality.

5.1.4 Context Hiding as a First Class security goal

The evolution of HS has established context hiding as a central security objective, transitioning from an optional privacy feature to an independent security dimension. Modern applications increasingly require that an evaluated signature reveal no information regarding the authenticated inputs, the intermediate values, or the specific structure of the computation beyond the final output. This maturation of the field is characterized by a move beyond weak context hiding toward complete context hiding, where an evaluated signature is mathematically indistinguishable from a fresh signature generated directly on the result. This ensures that even an adversary who has seen signatures on the original inputs cannot use the geometric structure of the evaluated object to infer the underlying context or the identities of the messages involved.

Achieving these rigorous privacy levels is particularly challenging in purely algebraic lattice-based schemes, where the signature's structure often reflects the linear relations of the computation. Consequently, a prominent trend is the shift toward modular architectures that utilize commitment schemes and zero-knowledge proofs. In these constructions, signers authenticate commitments rather than raw messages, and the evaluator provides a proof that the committed output corresponds to a valid computation over the committed inputs. This decoupling of authentication from privacy allows for the enforcement of strong context hiding without compromising unforgeability. Although these mechanisms often introduce computational and communication overhead, the growing demand for privacy-preserving data processing has made the integration of these fine-grained privacy guarantees a primary criterion in the design of next-generation homomorphic signature schemes.

5.1.5 Post Quantum Homomorphic Signatures

The shift toward post-quantum security is defined by the dominance of lattice-based constructions, which have moved from theoretical frameworks to standardized foundations for authenticated computation. Modern research has moved beyond the simple adaptation of number-theoretic primitives to leverage the intrinsic linear algebraic properties of the Short Integer Solution (SIS) and Learning With Errors (LWE) problems. These problems

provide a robust platform for homomorphic evaluation of linear functions and bounded-depth arithmetic circuits while offering worst-case hardness guarantees that are believed to be resilient against quantum adversaries. The current trend in this space is the optimization of Module-Lattice structures, which allow for a more efficient balance between signature size and computational overhead, facilitating the deployment of quantum-resistant signatures in resource-constrained environments like IoT and real-time cloud analytics.

A critical emerging frontier is the development of tightly secure and compact lattice-based signatures that maintain adaptive security without the massive efficiency penalties seen in early constructions. Recent advances focus on minimizing the growth of signatures during homomorphic evaluation through techniques like balanced rejection sampling and structured matrix trapdoors, which prevent the noise inherent in lattice operations from limiting the depth of admissible computations. Furthermore, the field is expanding into multi-key and leakage-resilient settings, where post-quantum signatures are designed to withstand active insider corruption and partial key exposure. This transition toward crypto-agile and scalable lattice architectures ensures that HS can provide not only long-term quantum resistance but also the functional flexibility required for complex, multi-tenant authenticated systems.

5.2 Open Problems

Despite a lot of progress over the last decade, HS are far from maturity. There are large number of challenges and open problems that persist, from efficiency, security modelling and practically.

Unifying Security Notions

The literature features a variety of security notions, including selective, semi-adaptive, adaptive unforgeability, chosen message versus chosen data attacks and different class of forgeries. However, developing a unified security framework that cleanly captures the nuances of multi-key environments, insider corruption and context-hiding simultaneously remains an open problem. Without this standardization, comparing the security of disparate constructions remains difficult, and the lack of concrete parameters prevents the development of cross-compatible cryptographic libraries for distributed systems.

Amortized Verification and Succinctness

While current Fully Homomorphic Signature schemes achieve communication succinctness (where the evaluated signature size is independent of the circuit), achieving computational succinctness remains a major challenge. In purely algebraic schemes, verification time typically scales linearly with the circuit size. A critical open research direction is the development of schemes that allow for sub-linear verification without the high overhead of general-purpose Zero-Knowledge proofs or the requirement for expensive, circuit-specific preprocessing. Specifically, the field seeks functional verification structures where the verifier's work is proportional only to the security parameter and the input/output dimensions, effectively decoupling the verification cost from the depth of the homomorphically evaluated circuit.

Post-Evaluation Extensibility and Re-evaluable Authentication

While modern multi-key homomorphic signature schemes allow signers to join adaptively without being declared at setup time, they typically bind authentication to the computation process itself. In particular, once a homomorphic evaluation has been performed, the resulting signature is often tightly coupled to the original set of signing keys and to proof objects generated during evaluation. A significant open problem is the design of

post-evaluation extensible MKHS schemes, in which additional parties can authenticate an already computed result without re-evaluating the circuit, regenerating zero-knowledge proofs, or incurring linear growth in signature size. Achieving such re-evaluable authentication would be especially relevant for decentralized settings such as collaborative data analysis or e-commerce platforms, where datasets evolve over time and participants may wish to endorse intermediate or final results retroactively.

Context hiding vs Efficiency

Existing homomorphic signature constructions reveal a fundamental tradeoff between context hiding and efficiency. Schemes that achieve weak or no context hiding typically rely on algebraic verification mechanisms, resulting in short signatures and fast verification, but at the cost of exposing the structure of the evaluated function or the coefficients used in the computation. In contrast, constructions providing strong or complete context hiding almost invariably rely on commitment schemes and non-interactive zero-knowledge proofs to hide the computation context, leading to significantly larger signatures and higher verification costs. At present, there is no known construction that achieves strong context hiding while retaining the efficiency and compactness of purely algebraic HS. Whether this tradeoff is inherent, or whether new algebraic or cryptographic techniques can reconcile strong context hiding with efficient verification, remains an important open question.

Homomorphic Signatures for Approximate Computation

Existing homomorphic signature schemes are designed for exact functional correctness, where verification requires that the output message matches the precise evaluation of a function over authenticated inputs. In contrast, modern approximate fully homomorphic encryption schemes, like CKKS [CKKS17] are only correct up to a bounded approximation error. Recent work on verifiable computation for approximate FHE [CCC⁺25] demonstrates that correctness can be certified within a tolerance, but does not address authenticity in the sense of HS. Bridging this gap raises fundamental challenges, including how to define unforgeability and correctness when valid outputs form equivalence classes rather than single values, and how to verify function evaluation without leaking information through approximation noise. Developing homomorphic signature schemes compatible with approximate computation, or hybrid frameworks combining authentication with verifiable approximate FHE, remains an open and largely unexplored research direction.

Finally, while lattice based HS provide post-quantum security, they are quite less efficient than pairing based schemes.

5.3 Conclusion

Homomorphic signatures have evolved from algebraically elegant authentication mechanisms for linear functions into a rich and diverse family of primitives supporting increasingly expressive computations, stronger security models, and decentralized trust assumptions. Early constructions, primarily based on bilinear pairings and later on lattices, demonstrated that homomorphic evaluation can be achieved by embedding authentication directly into algebraic structure. These schemes established the foundational paradigm of function-dependent verification, where correctness of a computation can be certified without access to the original data.

As the expressiveness of homomorphic signatures expanded, from linear and polynomial functions to leveled and fully homomorphic evaluation, it became increasingly clear that purely algebraic approaches face inherent limitations. In particular, achieving adaptive security, succinctness independent of circuit depth, and privacy notions such as context hiding often requires moving beyond direct algebraic manipulation. This insight has driven a

gradual but decisive shift toward proof-based constructions, where homomorphic evaluation is realized by producing verifiable proofs of correct computation over authenticated inputs. The integration of zero-knowledge arguments, functional commitments, and obfuscation has enabled fundamentally new capabilities, at the cost of heavier assumptions and more complex trust models.

The emergence of multi-key homomorphic signatures further highlights this transition. Unlike single-key schemes, multi-key settings require authenticating computations over data originating from mutually distrustful signers, often in the presence of insider corruption or dynamic participation. Modern MKHS constructions reveal that homomorphic authentication in decentralized environments is less about algebraic homomorphism and more about verifiable computation under distributed trust. Techniques such as proof aggregation, context binding, and re-evaluable authentication structures have become central, blurring the traditional boundary between homomorphic signatures, SNARKs, and verifiable computation frameworks.

Across all constructions, several fundamental trade-offs recur. Stronger privacy guarantees, such as context hiding, tend to conflict with succinctness and efficiency. Post-quantum security, while achievable through lattice-based designs, introduces challenges related to noise growth and correctness under repeated homomorphic evaluation. Likewise, robustness against key leakage and adaptive corruption often necessitates additional proof machinery, increasing system complexity. These tensions suggest that homomorphic signatures should no longer be viewed as a single primitive, but rather as a design space parameterized by trust assumptions, expressiveness, and deployment constraints.

Overall, the study of homomorphic signatures reflects a broader trend in modern cryptography: the convergence of algebraic authentication, zero-knowledge proofs, and verifiable computation. While significant progress has been made, particularly in multi-key and proof-based settings, many constructions remain heavy and far from practical deployment. Bridging this gap by simplifying proof systems, improving efficiency, and reconciling privacy with scalability, remains an open challenge. We hope that this SoK provides a unified perspective on the field, clarifies the conceptual relationships between existing approaches, and serves as a foundation for future work on practical, secure, and expressive homomorphic authentication.

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